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Breaking the Curse of Repulsion: Optimistic Distributionally Robust Policy Optimization

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Abstract

Off-policy policy optimization reuses historical behavior, including negative-advantage samples that suppress known failures. We show that repeated reuse can turn this useful signal into excessive repulsion. As the learner moves away from a historical negative action, subsequent negative updates make that action increasingly remote under the current policy while preserving or amplifying its influence. We formalize this far-field repulsive dynamic and show a transition from useful extrapolation to instability: moderate negative feedback can improve upon policies trained only on positive-advantage samples, whereas excessive far-field repulsion leads to persistent drift or the loss of finite stable equilibria. The resulting control problem is not merely distance weighting: learner-relative remoteness is nonlinear, corresponding to squared standardized distance for Gaussian policies and surprisal for categorical policies, while the local utility of remote negative feedback decays. Motivated by this principle, DRPO applies nonlinear remoteness-aware attenuation to retain informative near-field updates while suppressing far-field repulsion. Controlled experiments identify the causal mechanism, and D4RL locomotion and Countdown arithmetic generation show that the control remains beneficial in realistic external tasks.

1. Introduction

Off-policy policy optimization improves a current policy using behavior collected by earlier, stale, or heterogeneous policies. This paradigm is essential in settings where fresh interaction is costly, slow, or unavailable, including offline reinforcement learning, replay-based control, recommender systems, and the post-training of generative models. By reusing historical experience, it can reduce exploration cost, exploit data accumulated across policy iterations, and enable policy improvement without relying exclusively on new on-policy trajectories. This benefit, however, creates a distinctive learning problem: the policy continues to evolve while the data being optimized remain fixed. Understanding how this growing mismatch changes the influence of historical feedback is therefore central to building stable and effective off-policy learning algorithms.

Historical datasets, however, contain more than successful behavior to imitate. They also record suboptimal and failed actions that reveal which decisions should be suppressed. Policy-based methods exploit this distinction through signed feedback: positive updates increase the likelihood of successful behavior, whereas negative updates reduce the probability of known failures. Prior work has shown that such negative feedback can suppress undesirable modes, improve data efficiency, and benefit generalization in several policy-learning settings (Novati & Koumoutsakos, 2019; Fakoor et al., 2020; Arnal et al., 2025). Building on these observations, we view positive and negative feedback as complementary signals: positive updates provide attraction toward observed successes, while negative updates encode boundaries and failure information that positive imitation alone cannot express. We therefore ask how this information can be retained without allowing repeated historical reuse to turn negative feedback into excessive repulsion.

Against this background, we identify a self-amplifying far-field dynamic in repeated off-policy updates. A negative update pushes the learner away from a fixed historical action, increasing its learner-relative distance or surprisal. This increased remoteness can in turn strengthen—or prevent the decay of—the action's subsequent negative influence, causing repeated reuse to drive it progressively farther from the current policy. The precise amplification law depends on the policy family: continuous Gaussian policies can exhibit distance-dependent score growth, whereas categorical policies exhibit persistent suppression toward the support boundary. Thus, the danger lies not only in where a negative sample starts, but in how its influence evolves through repeated repulsion.

Prior work has stabilized off-policy learning through filtering, importance weighting, probability- or surprisal-aware

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gating, and other forms of tapered negative updates (Peng et al., 2019; Schulman et al., 2017; Fujimoto et al., 2019; Kumar et al., 2020; Arnal et al., 2025; Le Roux et al., 2025). These methods demonstrate the value of nonuniform control, but they do not isolate whether learner-relative remoteness itself generates a self-amplifying repulsive dynamic. In realistic data, sample quality, advantage magnitude, rarity, and policy distance are entangled. We disentangle these factors through matched controlled environments and causally test the pathway from remoteness, to amplified negative influence, to policy drift and collapse.

To characterize this process, we develop an aggregate theory of repulsive policy updates. The analysis shows that moderate negative feedback can improve upon what positive updates alone attain, whereas repeated far-field repulsion can progressively distort the aggregate equilibrium and ultimately eliminate a finite solution. This dynamic motivates DRPO, a learner-relative nonlinear control of negative updates. DRPO measures distance or surprisal under the evolving policy and applies an attenuation that is mild locally but increases nonlinearly in the far field. The resulting exponential envelope preserves informative local repulsion while dominating the far-field tail generated by the self-amplifying dynamics.

We evaluate the resulting theory and method through a layered empirical program. Controlled continuous and categorical experiments test the predicted transition from useful negative feedback to far-field instability and assess whether learner-relative nonlinear attenuation preserves the former while preventing the latter. We then examine whether the same mechanisms appear in continuous-control and sequence-generation tasks. Together, these experiments connect the proposed dynamics to controlled causal evidence, method behavior, and external task performance.

Our contributions are summarized as follows:

- we formulate repeated off-policy negative updates as learner-relative repulsive dynamics and develop an aggregate theory that characterizes stable extrapolation, boundary approach, and the loss of a finite equilibrium.
- we construct matched controlled environments and targeted near–far interventions that separate sample quality from learner-relative remoteness and causally trace amplified repulsion to policy drift and collapse.
- we propose DRPO, a learner-relative nonlinear attenuation of negative updates that uses distance or surprisal to preserve informative local feedback while suppressing the self-amplifying far-field tail.
- we evaluate the resulting theory and method across controlled continuous and categorical settings, together with external continuous-control and sequence-generation tasks.

2. Related Work

Learning from Negative or Suboptimal Behavior. A broad line of policy-learning methods distinguishes behavior according to its estimated quality. Advantage-weighted regression, AWAC, implicit Q-learning, critic-regularized regression, and behavior-proximal objectives fit actors more strongly toward actions estimated to be valuable under a learned critic (Peng et al., 2019; Nair et al., 2020; Kostrikov et al., 2021; Wang et al., 2020; Zhuang et al., 2023). Positive-filtered or filtering-based variants represent a conservative endpoint, where selected negative or low-quality updates are removed rather than reweighted. At the same time, failed or suboptimal behavior can remain informative: it can suppress competing bad modes, sharpen local decision boundaries, or improve data efficiency (Novati & Koumoutsakos, 2019; Fakoor et al., 2020; Arnal et al., 2025). These findings support the view that negative feedback is not merely noise, but also carries information whose usefulness depends on how it is used.

Off-Policy, Stale, and Learner-Relative Updates. This issue becomes especially important under off-policy reuse, where historical behavior remains fixed while the learner continues to change. Off-policy methods commonly control behavior–learner mismatch through importance weighting, clipping, trust regions, replay mechanisms, and behavior regularization (Schulman et al., 2017; Haarnoja et al., 2018; Levine et al., 2020; Fujimoto et al., 2019; Kumar et al., 2020). Recent asymmetric or tapered off-policy objectives further make updates depend on current-policy probability or surprisal (Arnal et al., 2025; Le Roux et al., 2025). These works establish that mismatch and rarity matter; our distinction concerns how learner-relative remoteness is created and amplified by repeated negative reuse. This actor-side view is complementary to conservative offline RL, which primarily addresses value extrapolation and policy-support mismatch.

3. Problem Setup

3.1. Off-Policy Signed Actor Updates

We consider policy optimization from a historical update distribution ν over state–action pairs, which may represent an offline dataset, a replay buffer, or trajectories collected by earlier or stale behavior policies. Let $\pi_\theta(a \mid s)$ denote the current policy and let $\widehat{A}(s, a)$ denote the advantage-like signal used by the actor update. During an actor step, the update distribution and the associated advantage estimates are treated as fixed with respect to θ. The resulting empirical actor field is

$$\mathbf{F}(\theta) = \mathbb{E}_{(s,a)\sim\nu}\left[\widehat{A}(s, a)\nabla_\theta \log \pi_\theta(a \mid s)\right], \qquad (1)$$

with the corresponding update $\theta_{t+1} = \theta_t + \eta \mathbf{F}(\theta_t)$. Equation (1) characterizes the actor-side dynamics induced by repeatedly reusing historical feedback. It is not assumed to equal the exact on-policy policy gradient, because the update distribution ν may differ from the state–action distribution of the current policy.

To separate favorable and unfavorable feedback, we define

$$\begin{aligned}\widehat{A}^{+}(s,a) &= \max\{\widehat{A}(s,a), 0\},\\ \widehat{A}^{-}(s,a) &= \max\{-\widehat{A}(s,a), 0\}.\end{aligned} \tag{2}$$

The actor field can then be decomposed into positive and negative components:

$$\begin{aligned}\mathbf{F}(\theta) = \mathbb{E}_{\nu}&\left[\widehat{A}^{+}(s,a)\nabla_\theta \log \pi_\theta(a \mid s)\right]\\ &- \mathbb{E}_{\nu}\left[\widehat{A}^{-}(s,a)\nabla_\theta \log \pi_\theta(a \mid s)\right].\end{aligned} \tag{3}$$

For a negative sample, the update magnitude factorizes as $\widehat{A}^{-}(s,a)\|\nabla_\theta \log \pi_\theta(a \mid s)\|$, separating the severity of its estimated disadvantage from its learner-relative policy geometry.

4. Far-Field Dynamics of Historical Policy Updates

We study how learner-relative remoteness shapes signed policy updates when historical behavior is reused. We first characterize, in policy-output coordinates, how the remoteness of a sample determines the strength of its score contribution. We then show how repeated reuse turns this static geometry into a dynamic feedback process: positive samples become self-attenuating as the learner approaches them, whereas negative samples become self-amplifying as the learner moves away. Finally, we analyze how these local effects aggregate into finite stable equilibria, persistent drift, or instability, and derive their specific manifestations in Gaussian and categorical policies.

4.1. Learner-Relative Remoteness and Update Strength

To isolate the action-side geometry, we fix a context s and write u for the policy-output coordinates at that context. For a historical action a, we define its learner-relative remoteness as

$$D_u(a) = -\log \pi_u(a). \tag{4}$$

For a categorical policy, $D_u(a)$ is the surprisal of the selected action. For a Gaussian policy, it equals a scale-dependent constant plus one half of the squared Mahalanobis distance from the policy mean. Thus, $D_u(a)$ measures how unlikely the historical action is under the current learner, rather than defining a metric in the action space.

We measure the corresponding squared score norm by

$$R_u(a) = \|\nabla_u \log \pi_u(a)\|_2^2. \tag{5}$$

A sample with advantage estimate $\widehat{A}(s,a)$ contributes an output-space update of magnitude $|\widehat{A}(s,a)|\sqrt{R_u(a)}$. The following results characterize how learner-relative remoteness controls this update strength.

Proposition 4.1 (Distance-dependent score-function magnitude)**.** *Fix a context and a historical action a.*

For a Gaussian policy with mean μ and fixed covariance $\Sigma \succ 0$, let

$$C_\Sigma = \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|\Sigma|. \tag{6}$$

The squared norm of the mean score function satisfies

$$\begin{aligned}R_\mu(a) &\geq 2\lambda_{\min}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big),\\ R_\mu(a) &\leq 2\lambda_{\max}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big).\end{aligned} \tag{7}$$

Writing $d^2_{\mathrm{Mah}}(a,\mu) = (a-\mu)^\top \Sigma^{-1}(a-\mu)$ for the squared standardized Mahalanobis distance, we have $D_\mu(a) - C_\Sigma = \frac{1}{2}d^2_{\mathrm{Mah}}(a,\mu)$. Hence, $R_\mu(a)$ grows quadratically with standardized distance and linearly with excess negative log-likelihood.

Moreover, along any fixed direction $v \neq 0$, with $a - \mu = rv$,

$$R_\mu(a) = 2\frac{v^\top \Sigma^{-2} v}{v^\top \Sigma^{-1} v}\big(D_\mu(a) - C_\Sigma\big). \tag{8}$$

For a categorical policy with logits z, the selected-logit score magnitude satisfies

$$\left|\frac{\partial}{\partial z_a}\log \pi_z(a)\right| = 1 - e^{-D_z(a)}. \tag{9}$$

It therefore increases strictly with surprisal but converges to the finite limit 1 as $D_z(a) \to \infty$.

Isotropic covariance. If $\Sigma = \sigma^2 I$, the directional coefficient is independent of v, and $R_\mu(a) = 2\big(D_\mu(a) - C_\sigma\big)/\sigma^2$. Thus, learner-relative remoteness and the squared mean score-function norm have the same global ordering.

A complete proof, together with bounds on the full categorical score-function norm, is provided in Appendix A.1.

Proposition 4.1 establishes the static distance–response relation underlying our analysis. For Gaussian policies, increasing remoteness along any fixed action-space direction produces an unbounded increase in the mean score-function norm, with a direction-dependent rate determined by the covariance. For categorical policies, selected-action suppression increases with surprisal but saturates at a finite ceiling. The next result shows how historical reuse turns these static relations into a dynamic feedback process.

4.2. Self-Amplification under Historical Reuse

The preceding result compares samples at a fixed policy. We now consider a fixed historical action that is repeatedly reused while the learner changes. Write

$$D(u) = D_u(a), \qquad R(u) = \|\nabla_u D(u)\|_2^2,$$

and consider the repeated single-sample update

$$u_{t+1} = u_t + \eta\widehat{A}\nabla_u \log \pi_{u_t}(a) = u_t - \eta\widehat{A}\nabla_u D(u_t). \quad (10)$$

Here, the action a and its signed coefficient $\widehat{A}$ are held fixed over the reuse window.

Theorem 4.2 (Self-attenuation and self-amplification under reuse)**.** *Assume that $D(u)$ is differentiable and convex along the update path, and define*

$$D_t = D(u_t), \qquad R_t = R(u_t).$$

If $\widehat{A} < 0$, then every reuse step satisfies

$$D_{t+1} \geq D_t + \eta|\widehat{A}|R_t, \qquad R_{t+1} \geq R_t. \quad (11)$$

Thus, negative reuse makes the historical action increasingly remote while preventing its score-function norm from decreasing.

If $\widehat{A} > 0$, and D has an L-Lipschitz gradient with $0 < \eta\widehat{A} \leq 1/L$, then

$$D_{t+1} \leq D_t - \frac{\eta\widehat{A}}{2}R_t, \qquad R_{t+1} \leq R_t. \quad (12)$$

Thus, positive reuse makes the historical action increasingly compatible with the learner while attenuating its subsequent score response.

A proof is provided in Appendix A.2. For the two policy-output coordinates used in the main theory, this convexity condition holds globally: the fixed-covariance Gaussian remoteness is quadratic in the mean, and the categorical negative log-likelihood is convex in logits. Theorem 4.2 distinguishes a one-time distant update from a historical-reuse feedback loop. A negative sample is first pushed farther from the learner; the resulting remoteness then preserves or increases its subsequent score response. Combined with Proposition 4.1, this produces unbounded amplification for fixed-covariance Gaussian mean updates and bounded but persistent amplification for categorical logits. These per-sample dynamics do not yet imply global divergence, because positive samples regain attractive force as the learner moves. We therefore next analyze their aggregate balance.

4.3. Aggregate Equilibria under Historical Reuse

The preceding results characterize the repeated influence of an individual historical action, but do not imply that the full policy must diverge: as negative repulsion moves the learner, positive samples regain attractive force. We therefore analyze the aggregate balance between positive and negative historical updates.

At a fixed context, consider a regular minimal exponential-family policy

$$\pi_\eta(a) = h(a)\exp\left\{\eta^\top T(a) - \psi(\eta)\right\}, \quad (13)$$

where $\eta \in \mathcal{H}$ is the natural parameter. Define the effective positive and negative update masses

$$p = \mathbb{E}_\nu\left[\widehat{A}^+(a)\right], \qquad q = \mathbb{E}_\nu\left[\widehat{A}^-(a)\right],$$
$$p\mathbf{m}_+ = \mathbb{E}_\nu\left[\widehat{A}^+(a)T(a)\right], \quad q\mathbf{m}_- = \mathbb{E}_\nu\left[\widehat{A}^-(a)T(a)\right]. \quad (14)$$

Thus, p and q combine sample frequency, advantage magnitude, and any fixed sample weighting. The aggregate signed actor field is

$$\mathbf{F}(\eta) = p\mathbf{m}_+ - q\mathbf{m}_- - (p-q)\nabla\psi(\eta). \quad (15)$$

Let

$$\mathcal{M} = \{\nabla\psi(\eta) : \eta \in \mathcal{H}\} \quad (16)$$

denote the interior mean-parameter space. We assume that ψ is twice continuously differentiable and strictly convex, and that $\nabla\psi : \mathcal{H} \to \mathcal{M}$ is one-to-one and onto.

Theorem 4.3 (Aggregate attraction–repulsion equilibria)**.** *Consider the continuous dynamics $\dot{\eta} = \mathbf{F}(\eta)$ and the discrete update $\eta_{t+1} = \eta_t + \alpha\mathbf{F}(\eta_t)$, with $\alpha > 0$.*

Positive-dominant regime $(p > q)$***.*** *Define the signed target*

$$\mathbf{m}^\star = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p-q}. \quad (17)$$

If $\mathbf{m}^\star \in \mathcal{M}$, there is a unique finite equilibrium $\eta^\star$ satisfying $\nabla\psi(\eta^\star) = \mathbf{m}^\star$. It is locally asymptotically stable under the continuous dynamics and locally stable under the discrete update for a sufficiently small step size. When $q > 0$,

$$\mathbf{m}^\star - \mathbf{m}_+ = \frac{q}{p-q}(\mathbf{m}_+ - \mathbf{m}_-), \quad (18)$$

so negative feedback moves the equilibrium beyond the Positive-only target in the direction away from the negative moment. As $\mathbf{m}^\star$ approaches the boundary of $\mathcal{M}$, the corresponding natural parameter leaves every compact subset of $\mathcal{H}$; if $\mathbf{m}^\star \notin \mathcal{M}$, no finite equilibrium exists.

Critical regime $(p = q)$***.*** *The field becomes the constant $p(\mathbf{m}_+ - \mathbf{m}_-)$. If $\mathbf{m}_+ \neq \mathbf{m}_-$, the policy exhibits persistent drift with no finite equilibrium. If $\mathbf{m}_+ = \mathbf{m}_-$, every point is stationary, but none is an isolated asymptotically stable equilibrium.*

Negative-dominant regime $(p < q)$***.*** *Any finite stationary point is unstable under both the continuous dynamics and every discrete update with* $\alpha > 0$*. Hence, this regime admits no locally asymptotically stable finite equilibrium.*

The derivation and proof, including the exact discrete-time step-size condition, are provided in Appendix A.3. Theorem 4.3 converts the preceding per-sample amplification mechanism into three terminal regimes: finite stable extrapolation, persistent drift, and instability. The extrapolation identity is geometric and does not by itself guarantee improved task utility. We next derive how these regimes appear in Gaussian and categorical policies.

4.4. Policy-Family Manifestations of Negative Dominance

Theorem 4.3 establishes that negative dominance, $p < q$, admits no finite stable equilibrium. It does not, however, determine whether the resulting runaway produces unbounded sample gradients. This depends on the policy family.

Theorem 4.4 (Unbounded Gaussian versus bounded categorical runaway)**.** *Assume* $p < q$ *and let* $r = q - p > 0$*.*

Gaussian policy. *Consider* $\pi_\mu = \mathcal{N}(\mu, \Sigma)$ *with fixed* $\Sigma \succ 0$*, and let* μ_+ *and* μ_- *denote the positive and negative weighted action means. Define*

$$\mu^\dagger = \frac{q\mu_- - p\mu_+}{q - p}. \tag{19}$$

The aggregate mean field is

$$\mathbf{F}_\mu(\mu) = r\Sigma^{-1}\left(\mu - \mu^\dagger\right). \tag{20}$$

Unless initialized exactly at the unstable stationary point $\mu^\dagger$*, both the continuous dynamics and every discrete update with step size* $\alpha > 0$ *satisfy* $\|\mu\|_2 \to \infty$*. Consequently, for every fixed historical action* a*,*

$$\|\nabla_\mu \log \pi_\mu(a)\|_2 = \left\|\Sigma^{-1}(a - \mu)\right\|_2 \longrightarrow \infty. \tag{21}$$

Categorical policy. *Consider* $\pi_z = \mathrm{softmax}(z)$ *with at least two actions, using the gauge* $\mathbf{1}^\top z = 0$*. Unless initialized exactly at a finite stationary point, when one exists, the continuous and discrete aggregate trajectories leave every compact subset of the logit space, and their probabilities approach the simplex boundary along a subsequence. Nevertheless, for every action* a *and every finite* z*,*

$$\begin{aligned}\|\nabla_z \log \pi_z(a)\|_2^2 &= \|e_a - \pi_z\|_2^2 \\ &\le 2\left(1 - \pi_z(a)\right)^2 \le 2.\end{aligned} \tag{22}$$

Thus, categorical runaway can produce unbounded logits and probability degeneration, but not an unbounded score-function norm.

The proof is provided in Appendix A.4. Theorem 4.4 reveals that the same aggregate instability has different policy-family manifestations. Gaussian mean runaway converts increasing action separation directly into unbounded far-field scores. Categorical policies instead approach the probability-simplex boundary while each sample score remains bounded. Hence, categorical degeneration is a bounded-gradient boundary event rather than Gaussian-style gradient explosion.

A learned Gaussian covariance is not required for the unbounded result. When covariance additionally contracts along a direction that remains separated from historical negative actions, it further amplifies the corresponding score. Neither policy-family result alone implies task performance collapse or a NaN/Inf numerical failure.

5. DRPO: Smooth Control of Far-Field Repulsion

Section 4 shows that negative feedback should not be controlled uniformly: nearby negative samples may contain useful local correction signals, whereas increasingly remote historical negatives can produce persistent or amplified repulsion. DRPO therefore controls only the negative branch with a learner-relative remoteness taper, aiming to retain local negative feedback while suppressing far-field influence. This section derives the minimum control order, motivates a stronger vanishing-tail requirement, and instantiates it with a smooth exponential taper.

5.1. Minimum-Order Control of Remoteness-Driven Expansion

We first ask how fast a negative-branch attenuation coefficient must decay to control the far-field expansion identified in Section 4. Let $r = d_{\mathrm{Mah}}(a, \mu)$ denote the standardized distance in the Gaussian mean geometry. The theory gives

$$D_\mu(a) - C_\Sigma = \frac{1}{2}r^2, \qquad R_\mu(a) = \|\nabla_\mu \log \pi_\mu(a)\|_2^2 \asymp r^2.$$

Thus, the raw far-field response is quadratic in standardized distance, and after attenuation its residual scale is $\omega(r)R_\mu(a)$. The minimum-order control requirement is the following.

Proposition 5.1. Minimum matching order. Suppose that the raw far-field response satisfies $I(r) = \Theta(r^2)$, and let $I_\omega(r) = \omega(r)I(r)$ be the residual response after applying a nonnegative attenuation coefficient $\omega(r)$. Then

$$I_\omega(r) = O(1) \quad \Longleftrightarrow \quad \omega(r) = O(r^{-2}) \qquad (r \to \infty).$$

Hence reciprocal-quadratic attenuation in standardized distance is the minimum matching order required to make the quadratic far-field response bounded. Any weaker attenuation cannot generally control the remoteness-driven expansion. The proof is given in Appendix A.5.

5.2. Distance-Dependent Utility of Negative Feedback

Section 5.1 gives a minimum-order condition for bounded far-field control. However, boundedness alone does not answer whether a remote historical negative update should continue to exert a nonzero residual force. We do not assume that negative feedback is intrinsically harmful. Rather, motivated by signed advantage updates in policy optimization, replay-discrepancy control, and gradient-alignment criteria for learning-signal usefulness (Novati & Koumoutsakos, 2019; Fakoor et al., 2020; Du et al., 2018), we introduce a learner-relative notion of negative-update utility.

Operationally, for a negative sample $z = (s, a)$, a weight $w \in [0, 1]$, and the raw repulsive update $g_\theta^-(z) = -\widehat{A}^-(z)\nabla_\theta \log \pi_\theta(a \mid s)$, we define its empirical utility on an evaluation batch $\mathcal{B}$ as

$$\widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta) = \widehat{J}_{\mathcal{B}}\big(\theta + \eta w g_\theta^-(z)\big) - \widehat{J}_{\mathcal{B}}(\theta). \tag{23}$$

This quantity measures the task effect of applying the weighted negative update under the current learner. We use it to state the design assumption underlying DRPO.

Design principle 5.2. Distance-dependent utility. The utility of a negative update is learner-relative and distance-dependent: nearby negative feedback may provide useful local boundary or correction information, whereas its utility should decrease as the sample becomes increasingly remote from the current policy. In the far field, once a historical negative update no longer provides useful local information, it should not retain a persistent nonzero repulsive force.

This principle strengthens the bounded-control floor in Section 5.1. Let $D_\theta(s, a)$ denote learner-relative remoteness, and let $I(D)$ denote the raw far-field influence scale expressed in the same remoteness coordinate. A merely bounded residual response can still assign a nonzero force to an unhelpful far-field sample. We therefore require the negative-branch attenuation $\omega(D)$ to satisfy

$$\omega(D)I(D) \longrightarrow 0 \qquad \text{as } D \longrightarrow \infty. \tag{24}$$

Thus, DRPO should preserve nearby negative feedback while driving far-field residual influence to zero. Appendix A.6 formalizes this utility view and gives sufficient conditions under which the utility-optimal far-field effective step vanishes.

5.3. Exponential Remoteness Taper

We now choose a concrete taper that satisfies the vanishing-residual requirement from Section 5.2. Since the far-field responses considered in Section 4 have finite growth order, a sufficient choice is an exponential envelope over learner-relative remoteness.

We instantiate DRPO with a thresholded exponential taper,

$$\omega_{\mathrm{Exp}}(D) = \exp\left\{-\lambda \left[\frac{D - \tau}{c}\right]_+\right\}, \tag{25}$$

where τ marks the onset of far-field control, $c > 0$ fixes the remoteness scale, and $\lambda > 0$ controls the decay rate. This taper leaves negative updates unchanged when $D \le \tau$, and attenuates them smoothly when $D > \tau$.

Proposition 5.3. Vanishing finite-order far-field influence. Let $I(D)$ denote the raw far-field influence scale. If, for some finite $m \ge 0$ and $C > 0$, $I(D) \le C(1 + D)^m$ in the far field, then the taper in Equation (25) satisfies

$$\omega_{\mathrm{Exp}}(D)I(D) \longrightarrow 0 \qquad \text{as } D \to \infty.$$

Thus, the exponential taper eliminates any finite-order far-field tail rather than merely bounding it.

This result provides the asymptotic guarantee required by Section 5.2: DRPO preserves nearby negative feedback while driving sufficiently remote residual influence to zero. The exponential form is not unique; it is a simple smooth envelope that dominates finite-order growth. The proof is given in Appendix A.7.

5.4. DRPO Update and Policy-Family Instantiations

We now embed the exponential remoteness taper into the signed actor update. Using the learner-relative remoteness $D_\theta(s, a) = -\log \pi_\theta(a \mid s)$, define $\bar{D}_\theta(s, a) = \mathrm{sg}[D_\theta(s, a)]$, where $\mathrm{sg}[\cdot]$ denotes stop-gradient. DRPO keeps the positive branch unchanged and attenuates only the negative branch:

$$\widetilde{A}_\theta(s, a) = \widehat{A}^+(s, a) - \omega_{\mathrm{Exp}}\big(\bar{D}_\theta(s, a)\big)\, \widehat{A}^-(s, a). \tag{26}$$

The resulting actor field is

$$\mathbf{F}_{\mathrm{DRPO}}(\theta) = \mathbb{E}_{(s,a)\sim\nu}\left[\widetilde{A}_\theta(s, a)\nabla_\theta \log \pi_\theta(a \mid s)\right]. \tag{27}$$

The remoteness statistic is recomputed under the current policy but detached before being used as a sample coefficient. Thus, DRPO is a learner-relative weighting of historical negative feedback rather than a differentiable distance regularizer.

Gaussian policies. For a Gaussian policy with mean $\mu_\theta(s)$ and fixed covariance Σ, define the squared standardized distance

$$\begin{aligned} D_{\mathrm{G},\theta}(s, a) &= d^2_{\mathrm{Mah}}(a, \mu_\theta(s)) \\ &= (a - \mu_\theta(s))^\top \Sigma^{-1}(a - \mu_\theta(s)). \end{aligned} \tag{28}$$

Since $D_\theta(s,a) - C_\Sigma = \frac{1}{2} D_{G,\theta}(s,a)$, absorbing the additive constant and the factor $1/2$ into the threshold and scale gives the Gaussian DRPO weight

$$\omega_G(s,a) = \exp\left\{-\lambda_G \left[\frac{D_{G,\theta}(s,a) - \tau_G}{c_G}\right]_+\right\}. \tag{29}$$

Categorical policies. For a categorical policy, learner-relative remoteness is the surprisal of the selected action, $D_\theta(s,a) = -\log \pi_\theta(a \mid s)$. Substituting this into Equation (25) gives

$$\omega_C(s,a) = \min\left\{1,\, (e^\tau \pi_\theta(a \mid s))^{\lambda/c}\right\}. \tag{30}$$

Thus, the same DRPO update becomes squared-standardized-distance control for Gaussian policies and surprisal control for categorical policies.

5.5. DRPO in the Space of Negative-Update Controls

The DRPO update in Equation (27) places negative feedback controls into a common form: positive updates keep unit weight, while the negative branch is controlled by a coefficient $\omega(D)$. This framing connects to advantage-weighted and positive-filtered policy learning, as well as tapered or probability-aware off-policy updates (Peng et al., 2019; Nair et al., 2020; Kostrikov et al., 2021; Grigsby & Qi, 2021; Arnal et al., 2025; Le Roux et al., 2025). The key distinction here is whether this coefficient depends on learner-relative remoteness.

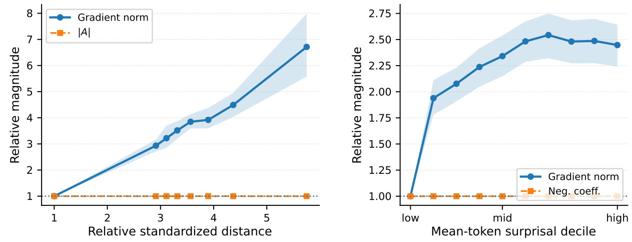
Distance-agnostic controls apply the same rule to all negative samples, whether they preserve, globally downweight, or remove the negative branch. They are useful baselines, but they cannot separate local negative feedback from the far-field tail: reducing one necessarily reduces the other.

Remoteness-aware controls instead make $\omega(D)$ decrease with distance. Their main difference is tail behavior. Reciprocal tapers provide polynomial decay and can yield bounded residual influence at the matching order, whereas exponential tapering decays smoothly and dominates finite-order far-field growth.

DRPO therefore occupies the regime targeted by the preceding analysis: it preserves local negative feedback while removing persistent far-field residual influence.

6. Experiments

Our experiments follow an outside–in–outside validation strategy. We first examine whether far-field negative influence arises in realistic continuous-control and autoregressive policy-optimization tasks. We then move to controlled environments that isolate learner-relative remoteness from sample quality, identify its contribution to negative-gradient amplification, and test whether this influence causally propagates into policy drift, support or variance boundary events, and task-performance collapse. Next, controlled taper experiments determine whether remoteness-dependent attenuation can preserve useful near-field repulsion while suppressing the harmful far-field tail. Finally, we return to external tasks to evaluate whether the same control principle transfers to complex policies and improves overall task performance.



(a) Hopper (b) Countdown

Figure 1. **External diagnostic of far-field negative influence.** **(a)** Hopper: relative implemented actor-gradient diagnostic and matched absolute advantage as functions of relative standardized distance for transitions with negative learned-critic advantages. Values are normalized to the near-field reference within each seed. Lines report averages across 10 seeds, and shaded regions denote seed-level bootstrap 95% confidence intervals. **(b)** Countdown: relative implemented actor-gradient diagnostic and fixed negative coefficient magnitude as functions of current-policy mean-token surprisal deciles among verifier-matched unsuccessful responses. Values are normalized to the lowest-surprisal decile, and shaded regions denote puzzle-level bootstrap 95% confidence intervals.

6.1. Environments and Datasets

We use two controlled environments and two external task domains. C-U1 is a contextual bandit with a Gaussian policy over continuous actions, while D-U1 uses unordered categorical actions with shared representations. They provide known ground truth for isolating far-field influence, policy dynamics, and held-out-context generalization. For external validation, we use D4RL Hopper and Countdown arithmetic generation, covering continuous offline control and autoregressive sequence policies, respectively. Full configurations are provided in Appendix B. We use policy-score norm for mechanism-source claims whenever the policy coordinate is controlled. In neural external diagnostics, we report implemented actor-gradient diagnostics only to check whether the policy-level remoteness effect appears in actual trainable updates.

6.2. External Evidence of Far-Field Repulsion

We first examine two external settings that instantiate learner-relative remoteness in different action spaces. Hopper provides a continuous-control setting with a state-conditioned Gaussian policy, where remoteness is measured by standardized distance from the current policy. Countdown provides an autoregressive discrete setting, where remoteness is measured by current-policy mean-token surprisal.

The diagnostic controls the sample-quality coefficient in each setting. In Hopper, we compare negative-advantage transitions after matching their absolute advantage magnitudes. In Countdown, we compare verifier-matched unsuccessful responses while holding the negative coefficient magnitude fixed. With the sample-quality coefficient controlled, these diagnostics test whether learner-relative remoteness is reflected in the implemented actor updates.

Figure 1 shows the same qualitative pattern in both domains. In Hopper, the implemented actor-gradient diagnostic increases with standardized distance while matched absolute advantage remains approximately flat. In Countdown, the implemented trainable-gradient diagnostic increases with mean-token surprisal while the negative coefficient remains fixed. increases with mean-token surprisal, while the negative coefficient remains fixed. These external diagnostics indicate that remoteness-correlated negative update influence appears in realistic neural actors. The controlled experiments below then isolate the policy-score source and test causal transmission.

The controlled experiments that follow isolate the source of this amplification and test whether it causally propagates into policy drift and collapse.

6.3. Controlled Identification of the Causal Mechanism

The external results show that far-field negative influence occurs in realistic tasks, but they are not designed to identify the causal mechanism. In Hopper and Countdown, sample quality, policy-relative remoteness, critic or verifier coefficients, data coverage, and subsequent policy dynamics remain partially entangled. We therefore turn to controlled environments, whose role is identification rather than realism: they allow sample quality and learner-relative remoteness to be varied independently, and allow near- and far-field negative updates to be intervened on directly. We first isolate whether remoteness itself changes negative influence and then test whether that influence causally propagates into policy drift and collapse. Full constructions and intervention protocols are provided in Appendix C.

6.3.1. Remoteness as an Independent Source of Negative Influence

We first isolate whether learner-relative remoteness changes negative influence independently of sample quality. In C-U1, negative actions are placed on the same reward contour, giving them identical fixed advantages but different standardized distances from the current Gaussian policy. The categorical direct-softmax reference applies the same factorization by fixing the negative coefficient while varying current-policy surprisal.

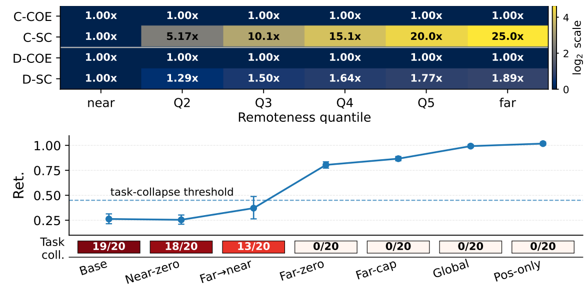


Figure 2. **Controlled source isolation and causal transmission. Top:** source-isolation factorization across learner-relative remoteness bins, normalized to the near bin. COE denotes coefficient magnitude and SC denotes policy-score response; C rows report the C-U1 continuous protocol, and D rows show the corresponding direct-softmax categorical reference. **Bottom:** C-U1 causal-intervention summary. Ret. denotes reward retention; points show 20-seed means with bootstrap confidence intervals, and Task coll. reports task-performance collapse counts.

The top panel of Figure 2 separates the coefficient term from the policy-score term. The coefficient magnitude remains flat across remoteness bins, whereas the score response increases with remoteness. Thus, the larger negative influence cannot be explained by farther samples having larger advantages; it is generated by the policy-score geometry itself. The manifestation differs across action spaces: Gaussian policies exhibit amplitude amplification, while categorical policies approach a bounded direct-logit score but continue to suppress the selected action toward the simplex boundary. This experiment identifies the source of the influence, not its downstream causal effect.

6.3.2. Causal Transmission into Drift and Collapse

We next test whether this far-field influence causally propagates into instability. In the C-U1 intervention protocol, all branches share the same data, fixed labels, initialization, optimizer, and training horizon, and differ only in how near- and far-field negative updates are retained, removed, capped, or globally rescaled.

The bottom panel of Figure 2 shows a clear rescue pattern. Removing near-field negative updates leaves performance collapse largely unchanged, whereas removing or capping far-field updates restores reward retention and eliminates task-performance collapse. Global rescaling also rescues the run, supporting the interpretation that abnormal negative-update magnitude is a mediator. In contrast, transferring the far-field budget to near-field negatives gives only partial recovery, showing that the rescue is not obtained by simply preserving the same total negative-update budget.

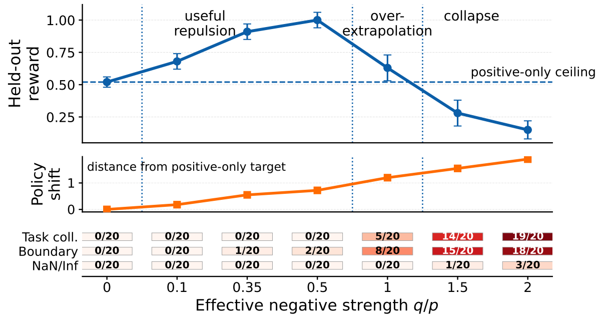
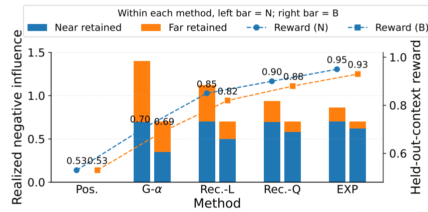


Figure 3. **Stable extrapolation from controlled negative feedback.** Increasing the effective negative strength q/p first improves held-out-context reward beyond the positive-only ceiling, then causes over-extrapolation and task-performance collapse. Policy shift measures displacement from the positive-only target. The color strip reports task collapse, boundary events, and NaN/Inf failures separately.

(a) Controlled taper comparison



(b) Transfer

Method	Hop.	Cnt.
Pos.	--	--
G-α	--	--
Rec.-L	--	--
Rec.-Q	--	--
EXP	**--**	**--**

Hop.: Hopper return. Cnt.: Countdown success.

Figure 4. **Controlled taper comparison and external transfer.** **(a)** Matched operating-point comparison in the controlled taper protocol. For each method, the left bar is the near-field-retention matched operating point and the right bar is the aggregate-budget matched operating point. Bars decompose retained negative influence into near- and far-field components, and dashed traces report held-out-context reward. **(b)** External validation summary on Hopper and Countdown. The right panel reports only final task-level outcomes; operating-point and matching details are shown in the controlled comparison.

These interventions identify far-field negative influence, rather than negative feedback in general, as a causal transmission path to policy instability in the controlled Gaussian environment. Task-performance collapse is visualized in the figure; support or variance-boundary events and NaN/Inf numerical failures are audited separately.

6.4. From Useful Repulsion to DRPO Control

The causal experiments show that the far-field tail must be controlled, but they do not imply that all negative feedback should be removed. The remaining question is how to preserve useful local repulsion while attenuating harmful remote influence. We therefore separate two roles: controlled taper-shape experiments identify which negative-weighting profiles preserve local utility under known near/far structure, whereas external experiments test whether the same control principle transfers to realistic offline-control and autoregressive-generation settings. Positive-only learning is stable but remains constrained by the observed positive support. C-U1 and D-U1 therefore place a hidden task optimum beyond the positive-only target and construct local negative feedback whose repulsive direction points toward that optimum, while additional remote negatives create far-field pressure. This exposes a controlled transition from useful local repulsion to harmful far-field dominance.

6.4.1. Stable Extrapolation from Controlled Negative Feedback

The causal interventions in Section 6.3.2 show that far-field negative influence must be controlled, but they do not imply that all negative feedback should be removed. We therefore sweep the effective negative strength relative to positive attraction, denoted by q/p, while keeping the data, labels, initialization, optimizer, and training horizon fixed; protocol and metric details are given in Appendix C.4. The positive-only setting is $q/p = 0$.

Figure 3 shows a non-monotone transition. For small and moderate q/p, held-out-context reward rises above the positive-only ceiling, showing that controlled negative feedback provides useful repulsion beyond the positive-support solution. As q/p grows larger, policy shift continues to increase but reward declines, indicating over-extrapolation rather than additional useful generalization. At the largest strengths, task-performance collapse and boundary events appear, while NaN/Inf failures are reported separately.

This strength sweep supplies the missing link between causal rescue and tapered control: positive-only learning removes the harmful tail but also removes the beneficial intermediate regime. Remoteness-aware control should therefore preserve useful local repulsion while preventing the far-field tail from dominating.

6.4.2. Controlled Comparison of Taper Shapes

We compare positive-only learning, global negative scaling, reciprocal-linear tapering, reciprocal-quadratic tapering, and exponential tapering while changing only the negative weighting rule. Two matched comparisons are used. Near-field-retention matching preserves comparable local negative influence and isolates differences in far-field decay, whereas aggregate-budget matching tests whether a selective taper outperforms a global reduction with the same total negative-update budget.

As shown in Figure **??**, exponential tapering provides the strongest stability–utility trade-off under both matched comparisons. Positive-only learning avoids far-field instability but remains at the positive-support ceiling. Global scaling suppresses useful near- and harmful far-field feedback indiscriminately. Reciprocal tapering introduces remoteness-aware suppression, but its tail remains heavier than the exponential rule. The exponential taper instead retains near-field repulsion while rapidly suppressing the far-field tail, yielding higher task performance and held-out-context generalization than the matched alternatives.

6.4.3. External Validation of Remoteness Tapering

We finally transfer the same control principle to Hopper and Countdown. DRPO weights Hopper transitions by their standardized remoteness from the current Gaussian policy and Countdown responses by their current mean token surprisal. Within each task, all methods share the dataset, initialization, labels, optimizer, and training budget. Because these external tasks do not provide known counterfactual geometry or exact near-field utility matching, they test transfer rather than repeat the controlled causal identification.

Table **??** shows that DRPO preserves the benefit of near-field negative feedback while avoiding the excessive displacement caused by insufficient far-field suppression. It achieves higher normalized return in Hopper and higher verifier success in Countdown than positive-only learning and global negative scaling, while producing fewer support or variance-boundary events. The controlled stability–utility principle therefore transfers to both continuous offline control and autoregressive policy optimization.

6.5. Overall Task Performance

Having established the mechanism and its transfer, we evaluate DRPO as a complete policy-optimization method on D4RL locomotion and Countdown. We compare it with published domain-specific baselines and protocol-matched controls spanning positive-only learning, global negative scaling, and alternative remoteness tapers. All protocol-matched methods use the same data and compute budgets. We report normalized return for D4RL and verifier success for Countdown, while task-performance failures, support or variance-boundary events, and NaN/Inf numerical failures remain separate outcomes.

6.5.1. D4RL Locomotion

We evaluate the nine D4RL Gym locomotion tasks formed by HalfCheetah, Hopper, and Walker2d under the medium, medium-replay, and medium-expert datasets. These datasets vary in behavior quality and coverage, with medium-replay containing the broadest mixture of historical policies. Table 1 compares DRPO with published offline-RL baselines under the D4RL evaluation protocol. The shared objectives and taper functions are defined in Appendix C.7; D4RL-specific score provenance, uncertainty statistics, remoteness construction, and implementation details are provided in Appendix C.8.

Table 1. Normalized returns on the D4RL-9 Gym locomotion benchmark.

Dataset	CQL†	TD3+BC†	IQL†	**DRPO**
halfcheetah-medium	44.0	**48.3**	<u>47.4</u>	47.3
hopper-medium	58.5	59.3	**66.3**	<u>63.3</u>
walker2d-medium	72.5	<u>83.7</u>	78.3	**85.5**
halfcheetah-replay	**45.5**	<u>44.6</u>	44.2	43.3
hopper-replay	<u>95.0</u>	60.9	94.7	**95.8**
walker2d-replay	77.2	**81.8**	73.9	<u>78.1</u>
halfcheetah-expert	**91.6**	90.7	86.7	<u>91.1</u>
hopper-expert	**105.4**	98.0	91.5	<u>101.8</u>
walker2d-expert	108.8	<u>110.1</u>	109.6	**110.4**
D4RL-9 total	<u>*698.5*</u>	*677.4*	*692.4*	***716.6***

Note. Dataset names are shortened for compactness: "replay" denotes "medium-replay", "expert" denotes "medium-expert", and the D4RL "-v2" suffix is omitted. Results marked by † are reported by prior work rather than protocol-matched reruns. Rows report task-level normalized returns, and *D4RL-9 total* sums the nine task means. Bold and underlined values denote the best and second-best results among the methods shown in this table. DRPO results are averaged over 10 runs; standard deviations and seed-level results are reported in Appendix C.8.

DRPO achieves the highest aggregate performance across D4RL-9, with a total normalized return of 716.6. Its advantage is largest on the medium-replay tasks, where broader behavioral coverage creates stronger far-field pressure, while remaining competitive on the medium and medium-expert datasets. These results indicate that selective remoteness-aware attenuation remains effective in a standard learned-critic offline-control benchmark.

6.5.2. Countdown

We evaluate DRPO on held-out Countdown puzzles, where a response is successful only if it forms a valid arithmetic expression that uses the provided numbers and reaches the target value. Greedy verifier success is the primary metric, while pass@k measures sampling-level solution coverage and valid-expression rate distinguishes reasoning failure from formatting or execution failure.

All compared policy-optimization methods start from the same SFT checkpoint and use the same frozen response bank, training budget, validation split, and test puzzles. We compare DRPO with AsymRE (Arnal et al., 2025) and TOPR (Le Roux et al., 2025) as off-policy baselines. The shared objectives and taper functions are defined in Appendix C.7; Countdown-specific baseline implementations, response-bank construction, checkpoint selection, and evaluation details are provided in Appendix C.9.

DRPO achieves the highest greedy verifier success among the methods shown in Table 2, reaching *[TBD]*. It improves over AsymRE and TOPR by *[TBD]* and *[TBD]*, respectively.

Table 2. Performance on held-out Countdown puzzles.

Metric	AsymRE	TOPR	**DRPO**
Greedy success (%)	–	–	–
Pass@k (%)	–	–	–
Valid expression (%)	–	–	–

Note. All methods use the same SFT checkpoint, frozen response bank, validation split, test puzzles, and training budget. A dash denotes a result pending the formal run.

DRPO also obtains the highest pass@k while maintaining a valid-expression rate of *[TBD]*, showing that the improvement comes from solving more puzzles rather than relaxing output validity.

7. Conclusion

This work reframes negative feedback in off-policy policy optimization as a controlled resource rather than a signal to be uniformly kept or removed. Its value depends on where a historical action lies relative to the current learner: nearby failures can shape useful decision boundaries, while remote failures can acquire disproportionate influence and destabilize optimization. DRPO operationalizes this view by attenuating negative influence according to learner-relative remoteness, preserving locally informative repulsion while limiting the far-field tail. The broader lesson is that stable offline and off-policy policy optimization should control the geometry of reused feedback, not merely its sign or magnitude.

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A. Proofs for the Theoretical Results

A.1. Proof of Proposition 4.1

Proof. Let $\delta = a - \mu$. For a Gaussian policy with fixed covariance Σ,

$$D_\mu(a) - C_\Sigma = \frac{1}{2}\delta^\top \Sigma^{-1}\delta. \tag{31}$$

The mean score function is

$$\mathbf{s}_\mu(a) = \nabla_\mu \log \pi_\mu(a) = \Sigma^{-1}\delta, \tag{32}$$

and hence

$$R_\mu(a) = \|\mathbf{s}_\mu(a)\|_2^2 = \delta^\top \Sigma^{-2}\delta. \tag{33}$$

Let $x = \Sigma^{-1/2}\delta$. Then

$$D_\mu(a) - C_\Sigma = \frac{1}{2}\|x\|_2^2, \qquad R_\mu(a) = x^\top \Sigma^{-1} x. \tag{34}$$

Applying the Rayleigh quotient bounds gives

$$\begin{aligned} R_\mu(a) &\geq 2\lambda_{\min}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big), \\ R_\mu(a) &\leq 2\lambda_{\max}(\Sigma^{-1})\big(D_\mu(a) - C_\Sigma\big). \end{aligned} \tag{35}$$

If $\delta = rv$ for a fixed direction $v \neq 0$, then

$$\begin{aligned} D_\mu(a) - C_\Sigma &= \frac{r^2}{2} v^\top \Sigma^{-1} v, \\ R_\mu(a) &= r^2 v^\top \Sigma^{-2} v. \end{aligned} \tag{36}$$

Eliminating r^2 proves Equation (8). For the categorical policy, let

$$p = \pi_z(a) = e^{-D_z(a)}. \tag{37}$$

The score function with respect to the logits is

$$\mathbf{s}_z(a) = \nabla_z \log \pi_z(a) = e_a - \pi_z. \tag{38}$$

Its selected-action component is therefore

$$\left|\frac{\partial}{\partial z_a} \log \pi_z(a)\right| = 1 - p = 1 - e^{-D_z(a)}, \tag{39}$$

which proves Equation (9). Since $1 - e^{-D}$ is strictly increasing for $D \geq 0$ and converges to 1 as $D \to \infty$, the selected-logit response increases with surprisal but remains bounded.

For completeness, the squared norm of the full categorical score function is

$$\begin{aligned} R_z(a) &= \|e_a - \pi_z\|_2^2 \\ &= (1-p)^2 + \sum_{j \neq a} \pi_z(j)^2. \end{aligned} \tag{40}$$

Because the non-selected probabilities sum to $1 - p$, the Cauchy–Schwarz inequality gives

$$\sum_{j \neq a} \pi_z(j)^2 \geq \frac{(1-p)^2}{K-1}. \tag{41}$$

The same sum is at most $(1-p)^2$. Hence,

$$\begin{aligned} \frac{K}{K-1}(1-p)^2 &\leq R_z(a) \\ &\leq 2(1-p)^2. \end{aligned} \tag{42}$$

Substituting $p = e^{-D_z(a)}$ completes the categorical part of the proof. $\square$

A.2. Proof of Theorem 4.2

Before proving the generic reuse result, we verify the convexity condition for the two policy-output coordinates used in the main text. For fixed-covariance Gaussian means,

$$D_\mu(a) = C_\Sigma + \frac{1}{2}(a-\mu)^\top \Sigma^{-1}(a-\mu), \qquad \nabla_\mu^2 D_\mu(a) = \Sigma^{-1} \succeq 0.$$

For categorical logits z, with $p = \mathrm{softmax}(z)$,

$$D_z(a) = \log \sum_j e^{z_j} - z_a, \qquad \nabla_z^2 D_z(a) = \mathrm{diag}(p) - pp^\top \succeq 0,$$

because

$$v^\top(\mathrm{diag}(p) - pp^\top)v = \mathrm{Var}_{i\sim p}(v_i) \geq 0.$$

Proof. Let

$$f(u) = D(u), \qquad g_t = \nabla f(u_t), \qquad R_t = \|g_t\|_2^2.$$

We first consider a negative coefficient $\widehat{A} = -c$ with $c > 0$. Defining $\alpha = \eta c$, the update becomes

$$u_{t+1} = u_t + \alpha g_t.$$

By convexity of f,

$$f(u_{t+1}) \geq f(u_t) + g_t^\top(u_{t+1} - u_t) \tag{43}$$
$$= f(u_t) + \alpha\|g_t\|_2^2. \tag{44}$$

Hence,

$$D_{t+1} \geq D_t + \eta|\widehat{A}|R_t.$$

The gradient of a differentiable convex function is monotone, so

$$\big(g_{t+1} - g_t\big)^\top \big(u_{t+1} - u_t\big) \geq 0.$$

Using $u_{t+1} - u_t = \alpha g_t$ gives

$$g_{t+1}^\top g_t \geq \|g_t\|_2^2.$$

By the Cauchy–Schwarz inequality,

$$\|g_{t+1}\|_2\|g_t\|_2 \geq g_{t+1}^\top g_t \geq \|g_t\|_2^2.$$

Whenever $\|g_t\|_2 > 0$, division by $\|g_t\|_2$ yields

$$\|g_{t+1}\|_2 \geq \|g_t\|_2,$$

and therefore $R_{t+1} \geq R_t$. The same result is immediate when $R_t = 0$. The inequality is strict whenever the gradient changes nontrivially along a direction of positive curvature.

We next consider a positive coefficient $\widehat{A} = c$ with $c > 0$. Let $\alpha = \eta c$, so that

$$u_{t+1} = u_t - \alpha g_t.$$

Because f has an L-Lipschitz gradient, the descent lemma gives

$$f(u_{t+1}) \leq f(u_t) - \alpha\|g_t\|_2^2 + \frac{L\alpha^2}{2}\|g_t\|_2^2 \tag{45}$$
$$= f(u_t) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|g_t\|_2^2. \tag{46}$$

If $\alpha \leq 1/L$, then

$$D_{t+1} \leq D_t - \frac{\alpha}{2} R_t = D_t - \frac{\eta \widehat{A}}{2} R_t.$$

It remains to show that the score-function norm does not increase. For a convex function with an L-Lipschitz gradient, the gradient is $1/L$-cocoercive:

$$\left(g_t - g_{t+1}\right)^\top \left(u_t - u_{t+1}\right) \geq \frac{1}{L} \|g_t - g_{t+1}\|_2^2.$$

Let $d = g_{t+1} - g_t$. Since $u_t - u_{t+1} = \alpha g_t$, this implies

$$d^\top g_t \leq -\frac{1}{\alpha L} \|d\|_2^2.$$

Consequently,

$$\|g_{t+1}\|_2^2 = \|g_t + d\|_2^2 \tag{47}$$

$$= \|g_t\|_2^2 + 2d^\top g_t + \|d\|_2^2 \tag{48}$$

$$\leq \|g_t\|_2^2 + \left(1 - \frac{2}{\alpha L}\right) \|d\|_2^2. \tag{49}$$

Because $\alpha L \leq 1$, the final coefficient is nonpositive, and hence

$$R_{t+1} = \|g_{t+1}\|_2^2 \leq \|g_t\|_2^2 = R_t.$$

This proves both claims. □

A.3. Derivation and Proof of Theorem 4.3

For the exponential-family policy in Equation (13), define the signed population objective

$$\mathcal{L}(\eta) = \mathbb{E}_{a \sim \nu} \left[\widehat{A}(a) \log \pi_\eta(a) \right]. \tag{50}$$

Using $\widehat{A} = \widehat{A}^+ - \widehat{A}^-$ and Equation (14), we obtain

$$\begin{aligned} \mathcal{L}(\eta) = &\left(p\mathbf{m}_+ - q\mathbf{m}_-\right)^\top \eta \\ &- (p - q)\psi(\eta) + C, \end{aligned} \tag{51}$$

where

$$C = \mathbb{E}_\nu \left[\widehat{A}(a) \log h(a) \right] \tag{52}$$

is independent of η. Therefore,

$$\nabla \mathcal{L}(\eta) = p\mathbf{m}_+ - q\mathbf{m}_- - (p - q)\nabla\psi(\eta) = \mathbf{F}(\eta), \tag{53}$$

and

$$\nabla^2 \mathcal{L}(\eta) = -(p - q)\nabla^2\psi(\eta). \tag{54}$$

Because the exponential family is regular and minimal, $\nabla^2\psi(\eta) \succ 0$ throughout $\mathcal{H}$.

Proof. **Positive-dominant regime.** Suppose $p > q$. A finite equilibrium must satisfy

$$\mathbf{F}(\eta^\star) = \mathbf{0}, \tag{55}$$

which is equivalent to

$$\nabla\psi(\eta^\star) = \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p - q} = \mathbf{m}^\star. \tag{56}$$

If $\mathbf{m}^\star \in \mathcal{M}$, existence follows because $\nabla\psi$ maps $\mathcal{H}$ onto $\mathcal{M}$, and uniqueness follows because this map is one-to-one.

Since $p - q > 0$, Equation (54) gives

$$\nabla^2 \mathcal{L}(\eta) \prec 0. \tag{57}$$

Thus, $\mathcal{L}$ is strictly concave, and $\eta^\star$ is its unique global maximizer.

For the continuous dynamics, the field Jacobian at the equilibrium is

$$J_{\mathbf{F}}(\eta^\star) = -(p-q)\nabla^2\psi(\eta^\star) \prec 0. \tag{58}$$

All eigenvalues are strictly negative, so $\eta^\star$ is locally asymptotically stable.

For the discrete update, the local update Jacobian is

$$J_{\text{disc}}(\eta^\star) = I - \alpha(p-q)\nabla^2\psi(\eta^\star). \tag{59}$$

Let $\lambda_i > 0$ denote the eigenvalues of $\nabla^2\psi(\eta^\star)$. The corresponding eigenvalues of $J_{\text{disc}}(\eta^\star)$ are

$$1 - \alpha(p-q)\lambda_i. \tag{60}$$

Their magnitudes are strictly smaller than one whenever

$$0 < \alpha < \frac{2}{(p-q)\lambda_{\max}\left(\nabla^2\psi(\eta^\star)\right)}. \tag{61}$$

This proves local discrete-time stability for sufficiently small α.

When $q > 0$, subtracting $\mathbf{m}_+$ from the signed target yields

$$\begin{aligned} \mathbf{m}^\star - \mathbf{m}_+ &= \frac{p\mathbf{m}_+ - q\mathbf{m}_-}{p-q} - \mathbf{m}_+ \\ &= \frac{q}{p-q}\left(\mathbf{m}_+ - \mathbf{m}_-\right). \end{aligned} \tag{62}$$

This proves Equation (18). If $q = 0$, the signed target reduces directly to $\mathbf{m}_+$.

If $\mathbf{m}^\star \notin \mathcal{M}$, no finite $\eta^\star \in \mathcal{H}$ can satisfy Equation (56), and hence no finite equilibrium exists.

Now consider a sequence $\mathbf{m}_n^\star \in \mathcal{M}$ that approaches the boundary of $\mathcal{M}$, and define

$$\eta_n^\star = (\nabla\psi)^{-1}\left(\mathbf{m}_n^\star\right). \tag{63}$$

Suppose, for contradiction, that $\{\eta_n^\star\}$ remains in a compact subset of $\mathcal{H}$. Then some subsequence converges to a finite $\bar{\eta} \in \mathcal{H}$. By continuity,

$$\mathbf{m}_n^\star = \nabla\psi(\eta_n^\star) \longrightarrow \nabla\psi(\bar{\eta}) \in \mathcal{M}, \tag{64}$$

contradicting convergence to the boundary. Hence, the corresponding natural parameters must leave every compact subset of $\mathcal{H}$.

Critical regime. Suppose $p = q$. Equation (53) reduces to

$$\mathbf{F}(\eta) = p\left(\mathbf{m}_+ - \mathbf{m}_-\right), \tag{65}$$

which is independent of η.

If $\mathbf{m}_+ \neq \mathbf{m}_-$, the continuous trajectory is

$$\eta(t) = \eta(0) + tp\left(\mathbf{m}_+ - \mathbf{m}_-\right), \tag{66}$$

and the discrete trajectory is

$$\eta_t = \eta_0 + t\alpha p\left(\mathbf{m}_+ - \mathbf{m}_-\right). \tag{67}$$

Both exhibit persistent linear drift and admit no finite equilibrium.

If $\mathbf{m}_+ = \mathbf{m}_-$, then $\mathbf{F}(\eta) = \mathbf{0}$ for every η. Every point is stationary, but a perturbed trajectory remains at the perturbed point rather than returning to the original one. Hence, no point is an isolated locally asymptotically stable equilibrium.

Negative-dominant regime. Suppose $p < q$. Equation (54) gives

$$\nabla^2 \mathcal{L}(\eta) = (q-p)\nabla^2 \psi(\eta) \succ 0. \tag{68}$$

Thus, $\mathcal{L}$ is strictly convex.

If a finite stationary point $\eta^\star$ exists, the continuous field Jacobian is

$$J_{\mathbf{F}}(\eta^\star) = (q-p)\nabla^2 \psi(\eta^\star) \succ 0. \tag{69}$$

All eigenvalues are positive, so the stationary point is repelling and therefore unstable.

For the discrete update, the local Jacobian is

$$J_{\mathrm{disc}}(\eta^\star) = I + \alpha(q-p)\nabla^2 \psi(\eta^\star). \tag{70}$$

Its eigenvalues are

$$1 + \alpha(q-p)\lambda_i > 1 \tag{71}$$

for every $\alpha > 0$ and every $\lambda_i > 0$. Therefore, any finite stationary point is unstable under the discrete dynamics as well. If no stationary point exists, there is trivially no finite stable equilibrium.

The three regimes establish Theorem 4.3. $\square$

A.4. Proof of Theorem 4.4

Proof. Let

$$r = q - p > 0. \tag{72}$$

Gaussian policy. For a Gaussian policy with fixed covariance,

$$\nabla_\mu \log \pi_\mu(a) = \Sigma^{-1}(a-\mu). \tag{73}$$

The aggregate mean field is therefore

$$\begin{aligned} \mathbf{F}_\mu(\mu) &= \mathbb{E}_\nu \left[\widehat{A}(a)\Sigma^{-1}(a-\mu) \right] \\ &= \Sigma^{-1} \left[p\mu_+ - q\mu_- - (p-q)\mu \right]. \end{aligned} \tag{74}$$

Because $p - q = -r$ and

$$\mu^\dagger = \frac{q\mu_- - p\mu_+}{r}, \tag{75}$$

Equation (74) becomes

$$\mathbf{F}_\mu(\mu) = r\Sigma^{-1}\left(\mu - \mu^\dagger\right). \tag{76}$$

For the continuous dynamics, define

$$\delta(t) = \mu(t) - \mu^\dagger. \tag{77}$$

Then

$$\dot{\delta}(t) = r\Sigma^{-1}\delta(t), \tag{78}$$

and hence

$$\delta(t) = \exp\left(r\Sigma^{-1}t\right)\delta(0). \tag{79}$$

Since $\Sigma^{-1} \succ 0$,

$$\|\delta(t)\|_2 \geq \exp\left(r\lambda_{\min}\left(\Sigma^{-1}\right)t\right)\|\delta(0)\|_2. \tag{80}$$

Therefore, unless $\delta(0) = 0$, the distance from the stationary point grows exponentially and $\|\mu(t)\|_2 \to \infty$.

For the discrete update,

$$\mu_{t+1} = \mu_t + \alpha \mathbf{F}_\mu(\mu_t), \tag{81}$$

so

$$\delta_{t+1} = \left(I + \alpha r \Sigma^{-1}\right) \delta_t. \tag{82}$$

Iterating gives

$$\delta_t = \left(I + \alpha r \Sigma^{-1}\right)^t \delta_0. \tag{83}$$

All eigenvalues of the update matrix are strictly larger than one. Consequently,

$$\|\delta_t\|_2 \geq \left[1 + \alpha r \lambda_{\min}\left(\Sigma^{-1}\right)\right]^t \|\delta_0\|_2 . \tag{84}$$

Thus, every nonstationary discrete trajectory also diverges.

For any fixed historical action a,

$$\begin{aligned} \|\nabla_\mu \log \pi_\mu(a)\|_2 &= \left\|\Sigma^{-1}(a-\mu)\right\|_2 \\ &\geq \lambda_{\min}\left(\Sigma^{-1}\right) \|a-\mu\|_2 . \end{aligned} \tag{85}$$

Since $\|\mu\|_2 \to \infty$ while a is fixed, $\|a-\mu\|_2 \to \infty$. Equation (85) therefore proves Equation (21).

Categorical policy. Let there be $K \geq 2$ actions and write

$$\pi_z = \mathrm{softmax}(z). \tag{86}$$

Because adding a constant to every logit does not change the policy, we work on the gauge-fixed subspace

$$\mathcal{H}_0 = \left\{z \in \mathbb{R}^K : \mathbf{1}^\top z = 0\right\}. \tag{87}$$

Let ρ_+ and ρ_- denote the normalized positive and negative weighted action distributions, and define

$$b = p\rho_+ - q\rho_-. \tag{88}$$

The signed categorical objective can be written as

$$\mathcal{L}(z) = b^\top z + r \log\left(\sum_{j=1}^{K} e^{z_j}\right) + C, \tag{89}$$

because $p - q = -r$. Its gradient and Hessian are

$$\nabla \mathcal{L}(z) = b + r\pi_z \tag{90}$$

and

$$\nabla^2 \mathcal{L}(z) = r\left[\mathrm{Diag}(\pi_z) - \pi_z \pi_z^\top\right]. \tag{91}$$

For every nonzero $v \in \mathcal{H}_0$,

$$\begin{aligned} v^\top \nabla^2 \mathcal{L}(z) v &= r\,\mathrm{Var}_{A\sim\pi_z}\left[v_A\right] \\ &> 0, \end{aligned} \tag{92}$$

because every softmax probability is positive and a nonzero vector in $\mathcal{H}_0$ cannot be constant. Hence, $\mathcal{L}$ is strictly convex on $\mathcal{H}_0$ and has at most one finite stationary point.

Consider first the continuous gradient flow

$$\dot{z} = \nabla \mathcal{L}(z). \tag{93}$$

Along every trajectory,

$$\frac{d}{dt}\mathcal{L}(z(t)) = \|\nabla \mathcal{L}(z(t))\|_2^2 \geq 0. \tag{94}$$

Suppose a nonstationary trajectory were bounded. Its closure would be compact, so $\mathcal{L}$ would remain bounded above. Integrating Equation (94) would then give

$$\int_0^\infty \|\nabla\mathcal{L}(z(t))\|_2^2 \, dt < \infty. \tag{95}$$

Since the gradient is Lipschitz on the compact trajectory closure, the standard gradient-flow argument implies

$$\nabla\mathcal{L}(z(t)) \longrightarrow 0. \tag{96}$$

Any accumulation point would therefore be a stationary point.

If no finite stationary point exists, this is an immediate contradiction. If a stationary point $z^\dagger$ exists, strict convexity gives, for every $z \neq z^\dagger$,

$$(z - z^\dagger)^\top \left[\nabla\mathcal{L}(z) - \nabla\mathcal{L}(z^\dagger)\right] > 0. \tag{97}$$

Since $\nabla\mathcal{L}(z^\dagger) = 0$,

$$\frac{d}{dt}\frac{1}{2}\left\|z(t) - z^\dagger\right\|_2^2 > 0 \tag{98}$$

along every nonstationary trajectory. Such a trajectory cannot have a subsequence converging to $z^\dagger$. Hence, the only bounded continuous trajectory is the stationary one itself.

For the discrete update

$$z_{t+1} = z_t + \alpha\nabla\mathcal{L}(z_t), \qquad \alpha > 0, \tag{99}$$

convexity gives

$$\begin{aligned} \mathcal{L}(z_{t+1}) &\geq \mathcal{L}(z_t) + \nabla\mathcal{L}(z_t)^\top (z_{t+1} - z_t) \\ &= \mathcal{L}(z_t) + \alpha \|\nabla\mathcal{L}(z_t)\|_2^2 . \end{aligned} \tag{100}$$

If the trajectory were bounded, Equation (100) would imply

$$\nabla\mathcal{L}(z_t) \longrightarrow 0. \tag{101}$$

A convergent subsequence would therefore approach a finite stationary point. If no such point exists, this is impossible. If the unique stationary point $z^\dagger$ exists, then for $z_t \neq z^\dagger$,

$$\begin{aligned} \left\|z_{t+1} - z^\dagger\right\|_2^2 &= \left\|z_t - z^\dagger\right\|_2^2 \\ &\quad + 2\alpha(z_t - z^\dagger)^\top \nabla\mathcal{L}(z_t) \\ &\quad + \alpha^2 \|\nabla\mathcal{L}(z_t)\|_2^2 \\ &> \left\|z_t - z^\dagger\right\|_2^2 . \end{aligned} \tag{102}$$

The trajectory therefore cannot have a subsequence converging to $z^\dagger$. Thus, every nonstationary discrete trajectory leaves every compact subset of $\mathcal{H}_0$.

To relate logit divergence to the probability boundary, note that on $\mathcal{H}_0$ the inverse softmax representation is

$$z_i = \log \pi_z(i) - \frac{1}{K}\sum_{j=1}^{K} \log \pi_z(j). \tag{103}$$

If the probabilities remained in a compact subset of the simplex interior, all coordinates would be bounded below by some $\varepsilon > 0$, and Equation (103) would imply bounded logits. Therefore, an unbounded gauge-fixed logit trajectory must contain a subsequence along which at least one action probability approaches zero.

Finally, for every action a,

$$\nabla_z \log \pi_z(a) = e_a - \pi_z. \tag{104}$$

Its squared norm satisfies

$$\begin{aligned}\|e_a - \pi_z\|_2^2 &= (1-\pi_z(a))^2 + \sum_{j\neq a} \pi_z(j)^2 \\ &\leq (1-\pi_z(a))^2 + \left(\sum_{j\neq a} \pi_z(j)\right)^2 \\ &= 2\,(1-\pi_z(a))^2 \\ &\leq 2.\end{aligned} \tag{105}$$

This proves the categorical score bound and completes the proof. □

A.5. Proof of Minimum-Order Control

Assume $I(r) = \Theta(r^2)$. Then there exist constants $0 < c_1 \leq c_2 < \infty$ and $r_0 > 0$ such that, for all $r \geq r_0$,

$$c_1 r^2 \leq I(r) \leq c_2 r^2.$$

The attenuated residual response is

$$I_\omega(r) = \omega(r) I(r).$$

If $I_\omega(r)$ is bounded, then for some $M < \infty$,

$$\omega(r) I(r) \leq M \qquad \text{for all sufficiently large } r.$$

Using $I(r) \geq c_1 r^2$, we obtain

$$\omega(r) \leq \frac{M}{c_1 r^2} = O(r^{-2}).$$

Thus bounded residual response requires reciprocal-quadratic decay.

Conversely, if $\omega(r) = O(r^{-2})$, then for some $C < \infty$,

$$\omega(r) \leq C r^{-2}$$

for all sufficiently large r. Using $I(r) \leq c_2 r^2$,

$$I_\omega(r) = \omega(r) I(r) \leq C r^{-2} \cdot c_2 r^2 = O(1).$$

Therefore reciprocal-quadratic attenuation is the minimum matching order needed to make the quadratic far-field response bounded.

A.6. Empirical Negative-Update Utility

To operationalize the utility view used in Section 5.2, let $\widehat{J}_{\mathcal{B}}(\theta)$ denote an empirical estimate of task performance on an evaluation batch $\mathcal{B}$. For a negative sample $z = (s, a)$, define its raw repulsive update as

$$g_\theta^-(z) = -\widehat{A}^-(z) \nabla_\theta \log \pi_\theta(a \mid s). \tag{106}$$

Assigning it a weight $w \in [0, 1]$ produces the parameter step

$$\Delta\theta(z; w) = \eta w g_\theta^-(z). \tag{107}$$

We define the empirical utility of this weighted negative update as

$$\widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta) = \widehat{J}_{\mathcal{B}}(\theta + \Delta\theta(z; w)) - \widehat{J}_{\mathcal{B}}(\theta). \tag{108}$$

The corresponding utility-optimal weight is

$$\widehat{w}^\star_{\mathcal{U}}(z \mid \theta) \in \arg\max_{w \in [0,1]} \widehat{\mathcal{U}}_{\mathcal{B}}(z; w \mid \theta). \tag{109}$$

We distinguish this empirical utility from its population counterpart:

$$\mathcal{U}_\theta(z;w) = J\big(\theta + \eta w g_\theta^-(z)\big) - J(\theta). \tag{110}$$

If $\widehat{J}_{\mathcal{B}}$ is an unbiased estimator of J, then

$$\mathbb{E}_{\mathcal{B}}\left[\widehat{\mathcal{U}}_{\mathcal{B}}(z;w \mid \theta)\right] = \mathcal{U}_\theta(z;w). \tag{111}$$

Thus, empirical utility is a noisy, evaluation-dependent estimate rather than an intrinsic deterministic label attached to the sample.

The following result gives sufficient conditions under which the utility-optimal effective step vanishes in the far field.

Lemma A.1 (Far-field shrinkage of the utility-optimal step)**.** *Let D index learner-relative remoteness, and let $g(D) \neq 0$ denote the corresponding negative-update direction. Write*

$$I(D) = \|g(D)\|_2, \qquad v(D) = \frac{g(D)}{I(D)}. \tag{112}$$

Assume that the local update space admits an orthogonal decomposition

$$\mathcal{T} = \mathcal{T}_{\text{task}} \oplus \mathcal{T}_{\text{far}}, \tag{113}$$

and let P_{task} denote the orthogonal projection onto $\mathcal{T}_{\text{task}}$. Suppose that:

1. *$\nabla J(\theta) \in \mathcal{T}_{\text{task}}$;*
2. *the normalized far-field update becomes asymptotically task-orthogonal,*
$$\|P_{\text{task}} v(D)\|_2 \longrightarrow 0; \tag{114}$$
3. *there exist $\rho_0 > 0$, $\kappa > 0$, and D_0 such that, for every $D \geq D_0$ and every $\rho \in [0, \rho_0]$,*
$$v(D)^\top \nabla^2 J(\theta + \rho v(D))\, v(D) \leq -\kappa. \tag{115}$$

Define the population utility of an effective step of length ρ by

$$\mathcal{V}_D(\rho) = J(\theta + \rho v(D)) - J(\theta), \qquad 0 \leq \rho \leq \rho_0, \tag{116}$$

and let

$$\rho^\star_{\mathcal{U}}(D) \in \arg\max_{\rho \in [0,\rho_0]} \mathcal{V}_D(\rho). \tag{117}$$

Then

$$\rho^\star_{\mathcal{U}}(D) \longrightarrow 0 \qquad \textit{as } D \longrightarrow \infty. \tag{118}$$

Moreover, if

$$\langle \nabla J(\theta), v(D) \rangle < 0,$$

then every nonzero local step has negative utility:

$$\mathcal{V}_D(\rho) < 0 \qquad \textit{for all } \rho \in (0, \rho_0]. \tag{119}$$

Proof. Define the first-order task alignment

$$a(D) = \langle \nabla J(\theta), v(D) \rangle. \tag{120}$$

Because $\nabla J(\theta) \in \mathcal{T}_{\text{task}}$, orthogonality gives

$$\begin{aligned} |a(D)| &= |\langle \nabla J(\theta), P_{\text{task}} v(D) \rangle| \\ &\leq \|\nabla J(\theta)\|_2 \, \|P_{\text{task}} v(D)\|_2 . \end{aligned} \tag{121}$$

Equation (114) therefore implies

$$a(D) \longrightarrow 0. \tag{122}$$

For any $\rho \in [0, \rho_0]$, Taylor's theorem gives some $\xi \in (0, \rho)$ such that

$$\begin{aligned}\mathcal{V}_D(\rho) &= \rho a(D) \\ &\quad + \frac{\rho^2}{2} v(D)^\top \nabla^2 J(\theta + \xi v(D))\, v(D).\end{aligned} \tag{123}$$

Using Equation (115),

$$\mathcal{V}_D(\rho) \le \rho a(D) - \frac{\kappa}{2}\rho^2. \tag{124}$$

Since $\mathcal{V}_D(0) = 0$, any utility-maximizing step must attain nonnegative utility. However, whenever

$$\rho > \frac{2[a(D)]_+}{\kappa}, \tag{125}$$

the right-hand side of Equation (124) is strictly negative. Hence,

$$0 \le \rho^\star_{\mathcal{U}}(D) \le \frac{2[a(D)]_+}{\kappa}. \tag{126}$$

Together with Equation (122), this proves Equation (118).

If $a(D) < 0$, then for every $\rho > 0$,

$$\rho a(D) - \frac{\kappa}{2}\rho^2 < 0. \tag{127}$$

Equation (124) therefore implies $\mathcal{V}_D(\rho) < 0$ for every nonzero local step, proving Equation (119). □

The lemma is a sufficient mechanism rather than a universal monotonicity claim. It applies when far-field negative updates become increasingly misaligned with the task-improving direction and the task objective is locally harmed along that far-field direction. The curvature condition can represent local losses caused by parameter drift or policy-support contraction, provided those effects reduce the evaluated objective J. If far-field growth remains aligned with $\nabla J(\theta)$, the conclusion need not hold.

Finally, the effective step generated by a sample weight w is

$$\rho(D; w) = \eta w I(D). \tag{128}$$

Consequently, the population analogue of the utility-optimal sample weight must satisfy

$$\eta w^\star_{\mathcal{U}}(D) I(D) = \rho^\star_{\mathcal{U}}(D) \longrightarrow 0. \tag{129}$$

Thus, when raw far-field influence remains nonvanishing or grows, the utility-optimal sample weight itself must shrink so that the weighted far-field influence vanishes.

A.7. Proof of Proposition ??

Proof. For $D > \tau$,

$$\omega_{\mathrm{Exp}}(D) = \exp\left[-\frac{\lambda}{c}(D - \tau)\right]. \tag{130}$$

Hence,

$$\begin{aligned}\omega_{\mathrm{Exp}}(D)\mathcal{I}(D) &\le C \exp\left[-\frac{\lambda}{c}(D-\tau)\right](1+D)^m \\ &\longrightarrow 0,\end{aligned} \tag{131}$$

because exponential decay dominates every finite-order polynomial.

The corresponding weighted parameter step is

$$\Delta\theta_{\mathrm{Exp}}(D) = \eta\omega_{\mathrm{Exp}}(D) g_\theta^-(D), \tag{132}$$

and therefore

$$\left\|\Delta\theta_{\mathrm{Exp}}(D)\right\|_2 = \eta\omega_{\mathrm{Exp}}(D)\mathcal{I}(D) \longrightarrow 0. \tag{133}$$

If $\widehat{J}_{\mathcal{B}}$ is locally $L_{\mathcal{B}}$-Lipschitz around θ, then for all sufficiently large D,

$$\begin{aligned}\left|\widehat{\mathcal{U}}_{\theta,\mathcal{B}}\big(z;\omega_{\mathrm{Exp}}(D)\big)\right| &\leq L_{\mathcal{B}}\left\|\Delta\theta_{\mathrm{Exp}}(D)\right\|_2 \\ &\longrightarrow 0.\end{aligned} \tag{134}$$

This proves both claims. □

B. Experimental Details

B.1. Unified Continuous Environment C-U1

C-U1 is the shared controlled continuous environment used by all continuous mechanism experiments. A context $s \in \mathbb{R}^6$ is sampled from $\mathcal{N}(0, I_6)$. For each seed, we independently sample 4096 training contexts and 4096 test contexts from the same distribution. Test contexts never participate in optimization and are used only to measure held-out-context generalization.

State-dependent task geometry. Each context determines a two-dimensional positive-support center $a_+(s)$ and a unit task direction $u(s)$. We define

$$\begin{aligned}[a_+(s)]_1 &= 0.70\tanh\big(0.85s_1 - 0.30s_2s_3 + 0.20\sin(1.6s_4)\big), \\ [a_+(s)]_2 &= 0.65\tanh\big(-0.50s_2 + 0.35\cos(1.1s_5) + 0.22s_1s_6\big).\end{aligned} \tag{135}$$

The direction angle is

$$\begin{aligned}\varphi(s) = {} & 1.15\tanh\big(0.75s_1 + 0.50s_3 - 0.30s_6\big) \\ & + 0.30\sin(1.35s_2),\end{aligned} \tag{136}$$

with

$$u(s) = \begin{bmatrix}\cos\varphi(s) \\ \sin\varphi(s)\end{bmatrix}, \qquad v(s) = \begin{bmatrix}-u_2(s) \\ u_1(s)\end{bmatrix}. \tag{137}$$

The unobserved task optimum and the directionally useful negative anchor are

$$\begin{aligned}a_\star(s) &= a_+(s) + 0.70u(s), \\ a_-(s) &= a_+(s) - 0.50u(s).\end{aligned} \tag{138}$$

Positive and negative actions. Each context is associated with four positive actions and eight negative actions. The positive actions lie on a reward contour of radius $r_+ = 0.75$ centered at $a_\star(s)$:

$$a_{\mathrm{pos},k}(s) = a_\star(s) + r_+\big(\cos\phi_k\, u(s) + \sin\phi_k\, v(s)\big), \tag{139}$$

where

$$\begin{aligned}\{\phi_k\}_{k=1}^4 &= \{\pi - \theta_1,\, \pi + \theta_1,\, \pi - \theta_2,\, \pi + \theta_2\}, \\ \theta_1 = 0.20, &\qquad \cos\theta_2 = \frac{2(0.70)}{0.75} - \cos\theta_1.\end{aligned} \tag{140}$$

This construction makes the four actions equal-reward while ensuring that their centroid is exactly $a_+(s)$. It also leaves nonzero conditional action spread, preventing deterministic maximum-likelihood variance contraction from being confused with the far-field mechanism.

The eight negative actions lie on a second reward contour of radius $r_- = 1.20$:

$$a_{\mathrm{neg},j}(s) = a_\star(s) + r_-\big(\cos\psi_j\, u(s) + \sin\psi_j\, v(s)\big), \tag{141}$$

with

$$\{\psi_j\}_{j=1}^{8} = \left\{ \pi,\ \frac{3\pi}{4},\ \frac{\pi}{2},\ \frac{\pi}{4},\ 0,\ -\frac{\pi}{4},\ -\frac{\pi}{2},\ -\frac{3\pi}{4} \right\}. \tag{142}$$

The first negative action equals $a_-(s)$. All eight negative actions have the same distance to $a_\star(s)$ and therefore the same reward and fixed advantage, while having different distances from the current policy.

Reward and fixed advantages. The ground-truth reward is

$$R(s, a) = \exp\left(-\frac{\|a - a_\star(s)\|_2^2}{2(0.75)^2} \right). \tag{143}$$

Advantages are computed once using the fixed baseline 0.40:

$$\widehat{A}(s, a) = R(s, a) - 0.40, \tag{144}$$

and remain frozen throughout optimization. Consequently, all four positive actions have positive advantages, all eight negative actions have negative advantages, and negative-action quality is exactly matched within each context.

Policy parameterization. The policy is an isotropic state-conditioned Gaussian,

$$\pi_\theta(a \mid s) = \mathcal{N}\left(\mu_\theta(s), \sigma_\theta^2(s) I_2\right). \tag{145}$$

Both $\mu_\theta(s)$ and the scalar $\log \sigma_\theta(s)$ are produced by a shared two-layer ReLU MLP with 64 hidden units per layer and separate output heads. The initial standard deviation is 0.60, and the learnable-variance experiments use no artificial variance clamp. The continuous mechanism results reported in the main text use the registered fixed-variance branch unless explicitly labeled otherwise; learnable-variance variants are labeled as audit ablations. A finite crossing of $\log \sigma_\theta(s) < -12$ is recorded as a support or variance-boundary event, whereas nonfinite parameters or outputs are reported separately as NaN/Inf numerical failure.

Training minibatches are sampled by context, and all actions associated with a selected context are retrieved together. Thus, the replicated actions within one context are not treated as independent contexts. Positive and negative objectives are averaged separately before their relative coefficient is applied, preventing the four-to-eight action count ratio from mechanically changing the total negative-update mass.

Shared use across continuous protocols. All C-U1 experiments reuse the same contexts, reward geometry, fixed advantages, and policy architecture. E1 compares equal-advantage negative actions at different learner-relative distances. E2 applies only positive updates and measures the same negative actions as phantom probes. E3 dynamically partitions negative actions using the standardized distance

$$d_\theta(s, a) = \frac{\|a - \mu_\theta(s)\|_2}{\sigma_\theta(s)}, \tag{146}$$

with the registered near–far threshold $d_\theta = 5.0$, and applies targeted near- and far-field interventions. E4 uses the aligned action $a_-(s)$ as directionally useful local negative feedback and the remaining contour actions as additional far-field pressure. The experiment-specific coefficients, training horizons, seeds, taper calibrations, and terminal criteria are reported separately with their corresponding protocols.

B.2. Controlled Categorical Environment D-U1

D-U1 denotes the controlled categorical environment family used to study persistent negative suppression, probability-support boundaries, and shared-semantic generalization. Its E6 instantiation uses a fixed catalogue of 64 unordered actions, each associated with a four-dimensional unit semantic vector. Action identifiers are randomly permuted for every seed, so numerical action indices carry no geometric ordering.

For each seed, we independently sample 2048 training contexts and 2048 test contexts from $s \sim \mathcal{N}(0, I_6)$. The two splits follow the same context distribution, and test contexts never participate in optimization. Results on this split are therefore reported as held-out-context or unseen-state generalization rather than OOD generalization.

Semantic action catalogue. Let $\{e_i\}_{i=1}^{K} \subset \mathbb{R}^4$, with $K = 64$, denote the unit reward embeddings of the categorical actions. They are sampled once per seed and randomly assigned to action identifiers. For each context, two fixed linear maps generate a demonstrated semantic target and a candidate improvement direction:

$$\begin{aligned} t_+(s) &= \operatorname{unit}\big(W_+^\top s\big)\,, \\ \widetilde{u}(s) &= W_u^\top s - \big\langle W_u^\top s, t_+(s)\big\rangle\, t_+(s), \\ u(s) &= \operatorname{unit}(\widetilde{u}(s))\,. \end{aligned} \tag{147}$$

The hidden optimal target and the local negative target are

$$\begin{aligned} t_\star(s) &= \operatorname{unit}(t_+(s) + \delta u(s))\,, \\ t_-(s) &= \operatorname{unit}(t_+(s) - \delta u(s))\,, \end{aligned} \qquad \delta = 0.45. \tag{148}$$

Reward and action roles. The ground-truth reward of action i is determined by its semantic alignment with the hidden target:

$$R(s,i) = \frac{1}{2}\left(1 + t_\star(s)^\top e_i\right). \tag{149}$$

The hidden optimal action is

$$i_\star(s) = \arg\max_i t_\star(s)^\top e_i. \tag{150}$$

It is excluded from all positive demonstrations.

For each context, the four positive actions are the available actions with the largest similarity to $t_+(s)$. After excluding the hidden optimum and positive actions, the local negative action is the action most aligned with $t_-(s)$. The four far-negative actions are selected by their alignment with $-t_+(s)$. Thus,

$$\begin{aligned} \mathcal{P}(s) &= \operatorname{Top4}_{i \neq i_\star(s)}\; t_+(s)^\top e_i, \\ i_{\text{local}}(s) &= \arg\max_{i \notin \{i_\star(s)\} \cup \mathcal{P}(s)} t_-(s)^\top e_i, \\ \mathcal{F}(s) &= \operatorname{Top4}_{i \notin \{i_\star(s), i_{\text{local}}(s)\} \cup \mathcal{P}(s)} \left[-t_+(s)^\top e_i\right]. \end{aligned} \tag{151}$$

The positive and negative advantages are fixed throughout training:

$$\widehat{A}(s,i) = \begin{cases} +1, & i \in \mathcal{P}(s), \\ -1, & i = i_{\text{local}}(s) \text{ or } i \in \mathcal{F}(s). \end{cases} \tag{152}$$

Consequently, differences between local and far-negative updates cannot be attributed to different advantage magnitudes.

Shared-semantic categorical policy. The policy uses a two-layer tanh MLP with 64 hidden units per layer to map the context to a unit semantic direction $q_\theta(s) \in \mathbb{R}^4$. Let $\widetilde{e}_i$ denote the action embedding used by the policy. The categorical logits and probabilities are

$$\begin{aligned} \ell_{\theta,i}(s) &= \kappa_\theta(s) q_\theta(s)^\top \widetilde{e}_i, \\ \pi_\theta(i \mid s) &= \frac{\exp(\ell_{\theta,i}(s))}{\sum_{j=1}^{K} \exp(\ell_{\theta,j}(s))}. \end{aligned} \tag{153}$$

The concentration is either fixed at $\kappa_\theta(s) = 8$ or learned as

$$\kappa_\theta(s) = \operatorname{softplus}(c_\theta(s)) + 0.05, \tag{154}$$

with initial value 8 and no upper clamp.

In the semantically aligned condition, $\widetilde{e}_i = e_i$. In the shuffled control, the same policy embeddings are randomly permuted across action identifiers while the reward embeddings and training samples remain unchanged. This intervention preserves the catalogue size and categorical parameterization but breaks the correspondence between policy-side similarity and ground-truth semantic utility.

Evaluation and failure events. We evaluate expected semantic reward, probability assigned to the hidden optimal action, entropy, effective support, and normalized semantic extrapolation on both training and held-out contexts. Task-performance collapse, effective-support or concentration-boundary events, and NaN/Inf numerical failure are reported separately. A policy can therefore retain high reward while still approaching a support boundary; the two outcomes are not treated as equivalent.

Shared use across categorical protocols. E6-A fixes the concentration and scans the strength of the local negative update to identify the transition from the positive-only ceiling to useful semantic extrapolation and then to performance reversal. E6-B learns the concentration and applies near/far causal interventions and matched controls. E6-C compares aligned and shuffled policy embeddings to determine whether improvements on the hidden action require the registered semantic correspondence.

E5 serves a distinct but complementary role. Its direct-softmax diagnostic and categorical causal reconstruction isolate persistent surprisal growth, logit-gap growth, and simplex-boundary suppression under repeated negative updates. It shares the fixed-advantage and near/far intervention logic of D-U1, but it does not use the 64-action shared-semantic catalogue and does not support the E6 semantic generalization claim. Its reconstruction parameters and direct-softmax initial conditions are reported separately in Appendix **??**.

B.3. D4RL Hopper Datasets

Hopper is a continuous-control task with an 11-dimensional observation space and a three-dimensional bounded action space. We use the D4RL `hopper-medium`, `hopper-medium-replay`, and `hopper-medium-expert` datasets. The medium dataset contains trajectories collected from a partially trained policy, medium-replay contains the corresponding replay buffer collected throughout training, and medium-expert combines medium and expert behavior. Hopper-medium-replay is used for the primary mechanism analysis because it provides broader behavioral coverage, while all three datasets are used for external method evaluation. Dataset versions, transition counts, and preprocessing statistics are reported in Table **??**.baselines.

B.4. Countdown Task and Puzzle Dataset

Countdown is an arithmetic generation task in which the policy receives a multiset of integers and a target value, and must generate an arithmetic expression that evaluates exactly to the target. A valid output must use every supplied number exactly once and may contain only addition, subtraction, multiplication, division, and parentheses.

Let $x = (\mathcal{N}, T)$ denote a puzzle, where $\mathcal{N}$ is the supplied multiset of numbers and T is the target. The policy action is a complete token sequence y representing the generated expression. Task success is determined by a deterministic verifier:

$$R(x, y) = \mathbf{1}\left[y \text{ is syntactically valid, uses } \mathcal{N} \text{ exactly once, and evaluates to } T\right]. \tag{155}$$

Because multiple expressions can solve the same puzzle, evaluation is based on verifier success rather than exact string matching.

The puzzle corpus is procedurally generated from valid arithmetic expressions and divided into disjoint training, validation, and test sets. Duplicate puzzles with the same number multiset and target are removed across the corpus. Validation puzzles are used for checkpoint selection, while the test split is reserved for final evaluation. Exact generation ranges, split sizes, and corpus statistics are reported in Table **??**.

Countdown provides an external discrete-action setting with autoregressive sequence generation and shared Transformer parameters. The construction of the offline response dataset, including positive, near-negative, and far-negative responses, belongs to the experimental protocol and is described separately in Appendix **??**.

B.5. Training and Evaluation Protocols

C. Controlled Experimental Protocols

This section specifies the interventions used in Sections 6.3 and 6.4. The definitions of C-U1, D-U1, Hopper, and Countdown are given in Appendix B. Network architectures, optimizer settings, training budgets, seeds, and evaluation frequencies are reported separately in Appendix **??**.

C.1. Shared Notation and Experimental Invariants

For a negative sample $z = (s, a)$, define its raw repulsive update as

$$g_\theta^-(z) = -\widehat{A}^-(z)\nabla_\theta \log \pi_\theta(a \mid s), \qquad \widehat{A}^-(z) = \max\{-\widehat{A}(z), 0\}. \tag{156}$$

Learner-relative remoteness is measured by

$$D_\theta(z) = -\log \pi_\theta(a \mid s), \tag{157}$$

or by its equivalent standardized Gaussian component when additive density constants are irrelevant.

Unless explicitly stated otherwise, all branches of one controlled comparison share:

1. the same training and evaluation contexts;
2. the same positive and negative samples;
3. the same fixed rewards or advantage labels;
4. the same initial policy parameters;
5. the same minibatch order, optimizer, and training horizon;
6. the same evaluation checkpoints and terminal classification rules.

Only the variable named by the intervention is changed. Actions generated under the same context are aggregated at the context level and are not treated as independent contexts for statistical inference.

C.2. Source-Isolation Protocol

Purpose. The source-isolation experiment asks whether policy-relative remoteness changes negative influence when sample quality is held fixed. It does not train a signed policy to collapse and therefore does not answer whether the resulting gradients cause downstream instability.

Continuous product-manifold construction. For each C-U1 context, we select multiple negative actions from a common reward contour:

$$R(s, a_i) = R(s, a_j), \qquad \widehat{A}(s, a_i) = \widehat{A}(s, a_j) < 0, \tag{158}$$

while ensuring that

$$D_\theta(s, a_i) \neq D_\theta(s, a_j). \tag{159}$$

The reward coordinate and the policy-distance coordinate are therefore independently controlled. This exact counterfactual comparison is the reason for using a synthetic product-manifold construction: an external offline dataset generally cannot provide two actions with precisely matched task quality and independently prescribed distance from the same current policy.

The comparison is performed both at policy initialization and after positive-only pretraining. No negative sample used as a probe is applied as an optimizer update during this protocol. Let $u_\theta(s)$ denote the policy-output coordinate used by the controlled probe, e.g., Gaussian mean or natural coordinates in C-U1 and direct logits in D-U1. For every probe, we record

$$\begin{aligned} I_{\text{score}}(z) &= \|\nabla_u \log \pi_u(a \mid s)\|_2 \,, \\ I_{\text{sample}}(z) &= |\widehat{A}(z)|\, I_{\text{score}}(z). \end{aligned} \tag{160}$$

For neural implementations, we additionally record implementation-level actor-gradient diagnostics and aggregate update norms as transfer checks, not as the theorem-level score measure. Because $|\widehat{A}|$ is identical across the matched actions, differences in I_{sample} arise from policy-score geometry rather than advantage magnitude. For every probe, we record

$$\begin{aligned} I_{\text{score}}(z) &= \|\nabla_\theta \log \pi_\theta(a \mid s)\|_2 \,, \\ I_{\text{sample}}(z) &= \left\|g_\theta^-(z)\right\|_2 \,, \end{aligned} \tag{161}$$

together with the full parameter-gradient norm and the aggregate gradient obtained from samples in the same remoteness range. Because $|\widehat{A}|$ is identical across the matched actions, differences in I_{sample} arise from policy score geometry rather than advantage magnitude.

We report near-to-far ratios of remoteness, score norm, single-sample gradient norm, aggregate gradient norm, and gradient-direction coherence. Ratios are first computed within each context and seed and are then aggregated across registered seeds.

Categorical counterpart. D-U1 uses unordered categorical actions with identical fixed negative labels. The comparison varies the current probability assigned to the negative action while preserving its reward role and advantage:

$$\widehat{A}(s,i) = \widehat{A}(s,j) < 0, \qquad \pi_\theta(i \mid s) \neq \pi_\theta(j \mid s). \tag{162}$$

For a direct categorical logit parameterization, the score norm is bounded and need not grow without limit as probability decreases. The relevant far-field effect is instead persistence: the negative update continues to increase the selected action's logit gap even after its probability has become small. We therefore report probability, surprisal, logit gap, score norm, and cumulative suppression over repeated updates. This distinguishes categorical support suppression from the unbounded Gaussian-distance amplification measured in C-U1.

C.3. Causal-Transmission Protocol

Purpose. The source-isolation protocol establishes where abnormal negative influence originates but does not establish that it changes task behavior. Causal transmission is therefore tested in the nonlinear Gaussian protocol, where the policy mean and variance evolve jointly, and in the categorical support reconstruction of D-U1.

Dynamic near–far partition. At each intervention step, negative samples are partitioned using their current-policy remoteness. Let $\mathcal{N}_\theta$ and $\mathcal{F}_\theta$ denote the registered near- and far-field sets. Their membership is recomputed as the policy evolves rather than frozen at initialization. The exact thresholds or quantiles are fixed before the formal run and reported in the implementation table.

Interventions. For a negative sample z, the following multipliers are applied to $g_\theta^-(z)$:

$$\begin{aligned} w_{\text{uncontrolled}}(z) &= 1, \\ w_{\text{positive}}(z) &= 0, \\ w_{\text{near-zero}}(z) &= \mathbf{1}[z \notin \mathcal{N}_\theta], \\ w_{\text{far-zero}}(z) &= \mathbf{1}[z \notin \mathcal{F}_\theta]. \end{aligned} \tag{163}$$

The far-cap intervention preserves the direction of a far-field update but limits its detached influence to a reference level estimated from near-field negative samples:

$$w_{\text{far-cap}}(z) = \begin{cases} \min\left\{1, \dfrac{C_{\text{near}}}{I_{\text{sample}}(z) + \epsilon}\right\}, & z \in \mathcal{F}_\theta, \\ 1, & \text{otherwise}, \end{cases} \tag{164}$$

where C_{near} is the registered near-field influence reference.

To distinguish selective intervention from a generic decrease in negative optimization, the global-matched control applies one scalar to every negative sample:

$$w_{\text{global}}(z) = \alpha_{\text{match}}. \tag{165}$$

The scalar is chosen using the training batch so that its detached aggregate negative influence matches the corresponding selective intervention:

$$\alpha_{\text{match}} \sum_{z \in \mathcal{B}^-} I_{\text{sample}}(z) = \sum_{z \in \mathcal{B}^-} w_{\text{selective}}(z) I_{\text{sample}}(z). \tag{166}$$

The matching coefficient is not differentiated through.

Causal contrasts. The primary causal contrast is between removing near-field and far-field negative updates. If Near-zero leaves the dynamics unchanged while Far-zero or Far-cap alters them, the effect cannot be attributed to negative feedback in general. The global-matched branch quantifies how much of the change follows from reducing total negative influence and how much requires selecting samples by remoteness.

Temporal and terminal measurements. We record policy displacement, reward, negative-gradient influence, mean dynamics and, for learnable-variance audit runs, scale dynamics, entropy or effective support, and the first time at which each registered event occurs. The analysis checks whether far-field influence rises before policy drift and whether drift precedes task-performance loss.

Terminal outcomes are classified separately as:

1. **task-performance collapse**: a sustained loss of the registered task metric while optimization remains finite;
2. **support or variance-boundary event**: categorical effective support or Gaussian variance crosses its registered boundary without requiring NaN or Inf;
3. **numerical collapse**: a parameter, loss, gradient, or model output becomes NaN or Inf.

A boundary event is not automatically counted as task-performance collapse, and finite execution is not automatically counted as convergence.

C.4. Controlled Negative-Strength Sweep

The controlled negative-strength sweep varies only the effective negative-repulsion strength q/p, where p denotes the aggregate positive-attraction budget and q denotes the aggregate negative-repulsion budget. All sweep arms share the same data, fixed labels, initialization, optimizer, training horizon, and event thresholds. The positive-only baseline corresponds to $q/p = 0$.

Held-out-context reward evaluates unseen-state generalization within the C-U1 controlled protocol. Policy shift is the normalized displacement of the learned policy from the positive-only target and visualizes how far negative feedback pushes the policy beyond the imitation solution. Task-performance collapse, support or variance-boundary events, and NaN/Inf numerical failures are audited as separate event types.

C.5. Controlled Taper Comparison

Purpose and controlled utility construction. Suppressing every negative update would remove the causal far-field pathway, but it would also remove potentially useful local repulsion. The taper comparison therefore requires an environment in which the utility of local negative feedback is known.

In C-U1, the hidden optimum $a_\star(s)$ lies beyond the center of the positive demonstrations $a_+(s)$. A local negative anchor is placed on the opposite side of $a_+(s)$, so repulsion from this anchor points toward $a_\star(s)$. Additional negative actions are placed on matched reward contours at larger policy-relative distances. Increasing negative influence therefore produces three observable regimes: positive-only under-extrapolation, useful local extrapolation, and far-field over-extrapolation or instability.

D-U1 implements the same logic with unordered semantic actions. The hidden optimal target lies beyond the demonstrated positive direction, while the local negative target lies in the opposite semantic direction. Repelling the local negative action can therefore improve the shared representation toward the hidden optimal action. Additional remote negative actions create support pressure without changing their fixed negative label. Evaluation on independently sampled contexts is reported as held-out-context generalization.

Weighting rules. For negative-sample remoteness D, the compared rules are

$$\begin{aligned}\omega_{\text{positive}}(D) &= 0,\\ \omega_{\text{uncontrolled}}(D) &= 1,\\ \omega_{\text{global}}(D) &= \alpha,\\ \omega_{\text{hard}}(D) &= \mathbf{1}[D \le \tau],\\ \omega_{\text{rec}}(D) &= \frac{1}{1+[(D-\tau)/c]_+},\\ \omega_{\exp}(D) &= \exp\left\{-\lambda\,[(D-\tau)/c]_+\right\}.\end{aligned} \tag{167}$$

All weighting values are computed from detached remoteness statistics; the policy cannot reduce its loss by differentiating through the taper.

Near-field-retention matching. Let $\mathcal{N}$ denote the registered useful near-field set and define its retained first-order influence as

$$\rho_{\text{near}}(\omega) = \frac{\sum_{z\in\mathcal{N}} \omega(D_\theta(z)) I_{\text{sample}}(z)}{\sum_{z\in\mathcal{N}} I_{\text{sample}}(z)}. \tag{168}$$

Taper parameters are calibrated on training contexts so that the compared selective rules retain the same registered near-field fraction, up to the predetermined tolerance. Parameters are then frozen before evaluation. This comparison isolates how the rules treat the far-field tail after preserving comparable local utility.

Aggregate-budget matching. For a weighting function ω, define its detached negative-update budget as

$$B(\omega) = \sum_{z\in\mathcal{B}^-} \omega(D_\theta(z)) I_{\text{sample}}(z). \tag{169}$$

The budget-matched global coefficient is

$$\alpha_{\text{budget}} = \frac{B(\omega)}{B(\omega_{\text{uncontrolled}})}. \tag{170}$$

This control applies the same total first-order negative influence without using remoteness. A selective taper can therefore be credited only for improvements that remain after comparison with its budget-matched global baseline.

Measurements. Each method is evaluated over the full registered training horizon. We report task performance, held-out-context generalization, distance from the hidden optimum, policy displacement, entropy or variance, effective support, near-field retention, aggregate negative-update budget, and all three terminal-event categories. Method comparisons use the same seeds and paired statistics.

C.6. External Taper Validation

Scientific scope. External taper experiments test whether the controlled weighting principle transfers to complex policies. They do not reproduce the exact counterfactual geometry of C-U1 or D-U1 and therefore do not replace the controlled source and causal experiments.

Hopper. All Hopper taper methods branch from the same actor initialization and use the same offline transitions and advantage labels. For an action with inverse-squash coordinate u, Gaussian mean $\mu_\theta(s)$, and diagonal scale $\sigma_\theta(s)$, the remoteness statistic is

$$D_{\text{Hopper}}(s,a) = \frac{1}{2}\sum_j \left(\frac{u_j - \mu_{\theta,j}(s)}{\sigma_{\theta,j}(s)}\right)^2. \tag{171}$$

The additive Gaussian normalization term is omitted because it does not change the ordering used by the taper. Remoteness is recomputed under the current actor and detached before constructing the negative weight. The mechanism protocol with frozen advantages is kept separate from the standard offline-RL performance protocol; the former studies transfer of the far-field control mechanism, whereas the latter reports final environment return.

Countdown. The Countdown offline dataset is generated once from a frozen SFT policy. For each puzzle, it contains a verified positive response, a near negative response with low mean token surprisal, and a far negative response with high mean token surprisal. Mean token surprisal is

$$D_{\text{Countdown}}(x, y) = -\frac{1}{|y|}\sum_{t=1}^{|y|} \log \pi_\theta \left(y_t \mid x, y_{<t}\right). \tag{172}$$

During method training, the responses remain fixed but their surprisal is recomputed under the current policy. The sequence-level taper weight is computed from the detached current surprisal and multiplies the negative sequence objective.

All Countdown methods start from the same SFT checkpoint and use the same positive, near-negative, and far-negative responses, mixture coefficients, validation set, and maximum training budget. Checkpoint selection uses the registered validation task metric, and final verifier success is measured only on the held-out test puzzles.

Comparisons and reporting. The common external comparisons are Positive-only, Uncontrolled, Global scaling, and exponential DRPO. Domain-specific baselines are reported in the overall-performance section rather than treated as controlled taper variants. Hopper reports normalized return and policy statistics; Countdown reports greedy verifier success, pass@k, valid expression rate, surprisal, and support statistics. Task-performance, boundary, and NaN/Inf outcomes remain separate in both domains.

C.7. Shared Objectives and Negative-Update Controls

This section defines the protocol controls and remoteness-aware weighting rules shared by the D4RL and Countdown experiments. Their task-specific remoteness signals, data construction, and optimization protocols are described separately in Appendices C.8 and C.9.

Shared signed objective. Let z denote an offline transition or response, let $A(z)$ denote its fixed signed learning signal during an actor update, and let $D_\theta(z)$ denote its learner-relative remoteness under the current policy. We write

$$A^+(z) = \max\{A(z), 0\}, \qquad A^-(z) = \min\{A(z), 0\}.$$

The shared family of actor objectives is

$$\mathcal{J}_\omega(\theta) = \mathbb{E}_{z\sim\mathcal{B}}\left[A^+(z)\log\pi_\theta(z) + \omega(D_\theta(z))\, A^-(z)\log\pi_\theta(z)\right],$$

where positive updates retain unit weight and $\omega(D)$ controls only the negative branch. During each actor update, the advantage and remoteness-derived weight are treated as fixed coefficients; gradients are not propagated through the weighting rule.

For the remoteness-aware methods, define the normalized excess remoteness

$$x_\theta(z) = \left[\frac{D_\theta(z) - \tau}{c}\right]_+,$$

where τ is the near-field threshold, $c > 0$ is the task-specific scale, and $[u]_+ = \max\{u, 0\}$. Consequently, samples inside the near-field region $D_\theta(z) \le \tau$ retain their full negative weight.

Positive-only. Positive-only removes all negative updates:

$$\omega_{\text{pos}}(D) = 0.$$

It therefore provides an extreme no-repulsion reference. In D4RL it retains only transitions with positive actor advantages; in Countdown it retains only verified successful responses under the corresponding binary learning signal.

Global-α. Global-α applies the same attenuation to every negative sample:

$$\omega_{\text{global}}(D) = \alpha, \qquad 0 \leq \alpha \leq 1.$$

This control tests whether performance improvements can be explained by a global reduction in negative-update magnitude, without using learner-relative remoteness.

Reciprocal-Linear and Reciprocal-Quadratic. The two reciprocal controls use the same remoteness signal and near-field threshold as DRPO, but differ in their far-field decay:

$$\omega_{\text{Rec-L}}(D) = \frac{1}{1 + \lambda x_\theta(z)},$$

and

$$\omega_{\text{Rec-Q}}(D) = \frac{1}{1 + \lambda x_\theta(z)^2}.$$

Reciprocal-Linear has the slowest far-field attenuation. The quadratic variant decays more rapidly but retains a polynomial tail. These controls isolate whether the benefit comes merely from using a remoteness signal or specifically from the decay profile selected by DRPO.

DRPO. DRPO uses an exponential remoteness taper:

$$\omega_{\text{DRPO}}(D) = \exp\left\{-\lambda x_\theta(z)\right\}.$$

The weight equals one throughout the near-field region and then decays exponentially once remoteness exceeds τ. Thus, DRPO preserves local negative feedback while suppressing the remote negative tail more rapidly than the reciprocal alternatives.

Hyperparameter selection. Within each task, τ, c, λ, and the Global-α coefficient are selected using the registered validation protocol only. Test-task performance is not used for hyperparameter selection. All protocol-matched methods share the same candidate budgets and checkpoint-selection rule. The task-specific search spaces and selected values are reported in Appendices C.8 and C.9.

C.8. D4RL Baselines and Experimental Protocol

The definitions of Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO follow Appendix C.7. This section specifies only the D4RL-specific baselines, remoteness construction, training protocol, and evaluation procedure.

Tasks and datasets. We evaluate HalfCheetah, Hopper, and Walker2d on the medium, medium-replay, and medium-expert datasets, producing the nine tasks reported in Table 1. The same fixed offline dataset is used by every protocol-matched method within a task.

Published offline-RL baselines. CQL, TD3+BC, IQL, and BC are included as published reference results. These values are not protocol-matched reruns and are therefore marked by † in Table 1. The main table reports their published task-level means. Their original uncertainty statistics and exact score provenance are recorded separately from the protocol-matched DRPO experiments.

Gaussian-policy remoteness. For an action $a \in \mathbb{R}^{d_a}$ and current Gaussian policy with mean $\mu_\theta(s)$ and standard deviation $\sigma_\theta(s)$, we use the following implementation-level standardized remoteness statistic

Table 3. Registered configuration for the D4RL performance experiments.

Configuration	Registered value
Actor-update budget	**[registered value]**
Evaluation interval	**[registered value]**
Evaluation episodes	**[registered value]**
Random seeds	**[registered seed set]**
Near-field threshold candidates, τ	**[registered grid]**
Remoteness-scale candidates, c	**[registered grid]**
Taper-strength candidates, λ	**[registered grid]**
Global-scaling candidates, α	**[registered grid]**

$$D_{\mathrm{D4RL}}(s,a) = \frac{1}{d_a} \sum_{j=1}^{d_a} \left(\frac{a_j - \mu_{\theta,j}(s)}{\sigma_{\theta,j}(s)} \right)^2 .$$

When the actor uses a transformed action distribution, this quantity is computed in the policy's native Gaussian coordinates. The resulting remoteness is used only to determine the negative-update coefficient; the distance term is detached during the actor update.

Protocol-matched actor comparison. Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO share the same offline dataset, critic checkpoint or critic-training protocol, actor initialization, network architecture, optimizer, minibatch schedule, actor-update budget, evaluation schedule, and random seeds. They differ only in the negative-update weighting rule defined in Appendix C.7.

The common actor advantage construction is held fixed across these methods. No method is allowed to change the critic, relabel samples, or alter the data distribution in order to improve its actor result.

Hyperparameters. The registered training and selection configuration is summarized in Table 3. All entries must be populated directly from the frozen experiment configuration rather than reconstructed after observing the test results.

These entries must be populated directly from the frozen experiment configuration rather than reconstructed after observing the test results.

Evaluation and uncertainty. Task performance is reported using the standard D4RL normalized return. For every protocol-matched method, the main table reports the mean over the registered seeds, while seed-level values and uncertainty statistics are reported in the appendix. The D4RL-9 total is the sum of the nine task-level means.

Task-performance collapse is assessed from normalized return and training trajectories. Variance or support-boundary events are reported separately from task performance. NaN/Inf numerical failures are reported as a third outcome and are not merged with either low return or policy-boundary events. Any claim concerning convergence, collapse, or method ranking is based on the registered terminal audit rather than an intermediate checkpoint.

C.9. Countdown Baselines and Experimental Protocol

Positive-only, Global-α, Reciprocal-Linear, Reciprocal-Quadratic, and DRPO follow the shared definitions in Appendix C.7. This section defines the two published off-policy baselines and the Countdown-specific response-bank, remoteness, training, and evaluation protocols.

AsymRE. Asymmetric REINFORCE defines the response-level advantage using a tunable reward baseline V:

$$A_{\mathrm{AsymRE}}(x,y) = r(x,y) - V,$$

and optimizes

$$\mathcal{J}_{\mathrm{AsymRE}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{B}}\left[\big(r(x,y) - V\big)\log \pi_\theta(y \mid x)\right].$$

For a binary verifier reward, lowering V emphasizes successful responses, whereas increasing V strengthens suppression of failed responses. The baseline is selected using the validation protocol and is held fixed within each run.

Global-α and AsymRE remain separate comparisons because they operate on different underlying learning signals. Global-α scales the negative part of the common precomputed signed advantage, whereas AsymRE constructs its signed advantage directly from the raw verifier reward and the published baseline parameterization. Under binary rewards and simplified normalization they are closely related, but the formal comparison preserves their native objectives and tuning rules.

TOPR. Let $\mu(y \mid x)$ denote the frozen response-generating policy and

$$\rho_\theta(x,y) = \frac{\pi_\theta(y \mid x)}{\mu(y \mid x)}$$

the sequence-level importance ratio. Let $R(x,y)$ denote the signed reward used by TOPR and define

$$\mathcal{T}^+ = \{(x,y) : R(x,y) \geq 0\}, \qquad \mathcal{T}^- = \{(x,y) : R(x,y) < 0\}.$$

We implement canonical TOPR, whose expected update is

$$\begin{aligned} g_{\mathrm{TOPR}}(\theta) = \mathbb{E}_{(x,y)\sim\mu}&\Big[\mathbf{1}\{(x,y)\in\mathcal{T}^+\}R(x,y)\nabla_\theta \log \pi_\theta(y \mid x)\Big] \\ &+ \mathbb{E}_{(x,y)\sim\mu}\Big[\mathbf{1}\{(x,y)\in\mathcal{T}^-\}\min\{\rho_\theta(x,y),1\}R(x,y)\nabla_\theta \log \pi_\theta(y \mid x)\Big]. \end{aligned}$$

Thus, positive responses receive the SFT-style unit coefficient, while negative responses are attenuated using a truncated importance ratio. The behavior-policy log probability required by the ratio is stored when the frozen response bank is constructed.

Countdown remoteness. For a prompt x and completion $y = (y_1, \ldots, y_T)$, learner-relative remoteness is the current mean completion-token surprisal

$$D_{\mathrm{CD}}(x,y) = -\frac{1}{T}\sum_{t=1}^{T} \log \pi_\theta\left(y_t \mid x, y_{<t}\right).$$

Only completion tokens are included; prompt tokens and padding positions are excluded. Mean rather than summed surprisal is used so that response length does not mechanically determine remoteness. The remoteness score is detached before applying the shared taper functions.

Frozen response-bank construction. The response bank is generated before policy optimization and then held fixed. Each record contains the puzzle, generated response, verifier outcome, response length, and the behavior-policy log probability required by methods that use importance ratios. The same bank and training split are supplied to every protocol-matched method.

A response is marked successful only when it forms a valid arithmetic expression, uses the supplied numbers according to the task rules, and evaluates to the target. Invalid syntax, illegal number use, and incorrect arithmetic results remain distinct verifier outcomes in the stored diagnostics, even when they share the same binary training reward.

Training and checkpoint selection. All methods start from the same SFT checkpoint and share the optimizer, learning-rate schedule, minibatch construction, number of optimizer steps, validation split, test puzzles, and random seeds. Hyperparameters are selected using the registered validation rule. The held-out test set is evaluated only after checkpoint selection.

Table 4. Registered configuration for the Countdown performance experiments.

Configuration	Registered value
Optimizer-step budget	**[registered value]**
Batch size	**[registered value]**
Learning-rate schedule	**[registered value]**
Random seeds	**[registered seed set]**
AsymRE baseline candidates, V	**[registered grid]**
Near-field threshold candidates, τ	**[registered grid]**
Remoteness-scale candidates, c	**[registered grid]**
Taper-strength candidates, λ	**[registered grid]**
Global-scaling candidates, α	**[registered grid]**
Number of samples for pass@k	**[registered value]**

The registered training and selection configuration is summarized in Table 4. All entries must be populated directly from the frozen experiment configuration rather than reconstructed after observing the test results.

Verifier-based evaluation. Greedy verifier success is the fraction of held-out puzzles solved by greedy decoding. Pass@k is the fraction solved by at least one of the registered k sampled responses. Valid-expression rate is the fraction of generated responses that satisfy the parser and number-use constraints, irrespective of whether they reach the target.

These three metrics are reported separately. In particular, a higher verifier success rate is not attributed to improved reasoning when it is accompanied by a deterioration in expression validity.

Task-performance degradation, token- or sequence-support boundary events, and NaN/Inf numerical failures are audited independently. Claims about stability or final method ranking use the registered terminal checkpoint audit rather than finite-step pilot behavior.